

HERMES HF Telecommunications Manual (Combined)

Rhizomatica

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HERMES

Abstract

This document is intended to be the primary documentation of the HERMES project. It aggregates the main HERMES manuals into a single, standard LaTeX source, so it remains easy to compile as PDF (pdf_{flat}ex) and to convert to HTML with tools such as `latex2html`.

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Chapter 1

User manual

1.1 Introduction

HERMES is a telecommunication system which operates in the High Frequency (HF) band. HERMES allows digital multimedia exchange between users and stations, including exchange of text, image, audio or any other file type. The system interface to the users is a web interface available to users within WiFi connectivity distance to a HERMES transceiver.

The system relies heavily on the e-mail protocol, which can be managed and accessed through the system's web interface or through specialized apps, like DeltaChat¹. HERMES also supports peer-to-peer secure messages between the Host stations, which works as a bulletin board system (BBS).

The system employs a star topology network in which a Gateway station connects to all Host stations in remote locations. The Gateway station routes e-mail and other messages locally (back to Host stations over HF) or over the Internet to the wider world. The synchronization between the data to be sent or received from each remote station is asynchronous and orchestrated by the Gateway station in a round-robin fashion (one Host station after another, in order) during pre-established times.



¹DeltChat, a multi-plataform e-mail messenger: <https://delta.chat/>

1.1.1 HERMES HF Transceiver

The front panel of the equipment is shown in Figure 1.2.

The HERMES box includes the following input / output interfaces, as shown in Figure 1.1.

1. Back Panel - Serial number;
2. Back Panel - Ground connector;
3. Back Panel - Ventilation openings;
4. Back Panel - HF antenna connector (PL-259 / UHF female);
5. Back Panel - Fuse (10A);
6. Back Panel - 12V DC power input, positive (red) and negative (black) terminals;
7. Back Panel - WiFi antenna connector (RP-SMA female);
8. Back Panel - RJ-45 Ethernet port, for connection to external switch or router;
9. Front Panel - Power switch (on/off);
10. Front Panel - 4 indicator LED's (System LED, Antenna Status LED, Connected LED, TX LED).

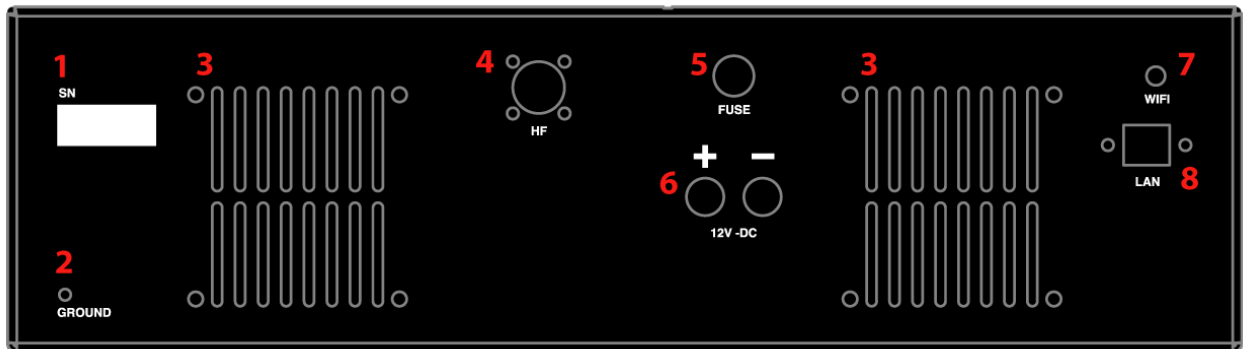


Figure 1.1: Back view of HERMES box

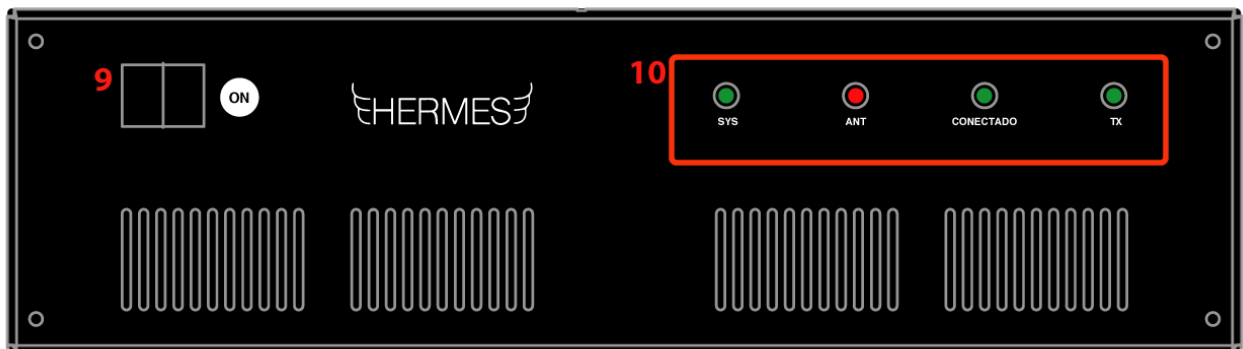


Figure 1.2: Indicator's LED on front panel

The LED indicator lights are set to show the following:

- SYS: System LED, blinks when the system is not ready to be used. SYS LED typically blinks when the equipment is turning on or when it is turning off;

- ANT: Green when the antenna is functioning properly, and no high SWR value is detected. ANT LED turns Red when a high SWR value is detected, meaning the high SWR protection is enabled and the radio will not transmit. SWR protection needs to be reset if active (Red), for the radio to become operational again, which can be done in the radio administrative interface. Check the RF cable and Antenna before resetting in the interface;
- CONECTADO: LED on (solid Green) means that the station is connected to another station;
- TX: LED on means the radio is transmitting. When this LED is off, it is in receive mode. If the CONECTADO LED is on and the TX LED is off, the station is actively receiving. During active communication with another station the CONECTADO LED will be solid Green and the TX LED will turn on and off;

1.1.2 HF Antenna Recommendation

An HF antenna tuned to the desired operating frequency must be connected to the equipment. Never turn on the equipment without first connecting to an appropriate antenna!

There are many HF antennas, each one fitting a different purpose. For short and medium range communications (up to about 800 km), a quarter wavelength dipole installed in inverted V configuration is a good and affordable option.

1.1.3 Power Requirements

The system is designed to operate with between 12V DC to 14V DC power. A typical setup would use 12V DC from a battery connected to a solar charge controller and photo-voltaic panels. Another setup option is to use a 12V or 13.8V AC/DC power supply.

The red connector on the back of the transceiver should be connected to the positive (+) polarity of the battery or power supply unit, while the black connector to the negative (-) polarity. The system has polarity inversion protection, but care should be taken to proper power installation of power connections. The transceiver consumes approximately 2A in receive mode, and 6A in transmit mode.

1.2 Web Interface

HERMES provides a web interface which can be accessed through a local WiFi network. When accessing the HERMES WiFi network, the following information applies:

- Network Name (ESSID): HERMES
- Password: amazonia

Be aware that on some mobile phones a browser will automatically open with the the main HERMES web page, while in others, to access the HERMES web interface, you must open the browser and access <http://ac1.hermes.radio> or <http://10.0.0.1>.

O navegador chrome / chromium é o navegador padrão para utilização da interface HERMES, para garantir uma melhor experiência faça preferência pelos navegadores informados.



Figure 1.3: Hermes home

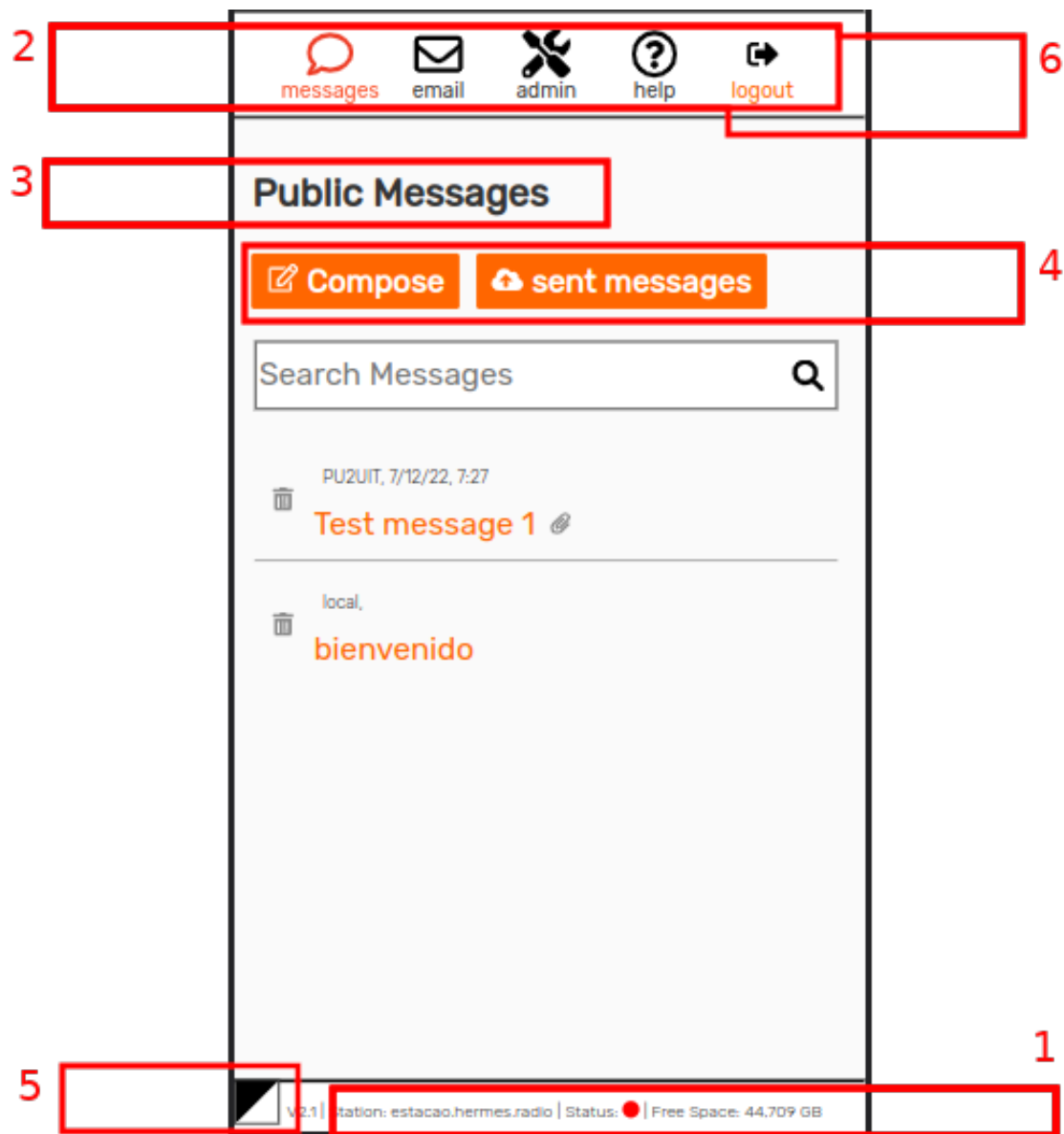


Figure 1.4: Hermes public messages and elements

On the public messages page 1.4 you will find the following links:

1. Station information and status;
2. Main menu;
3. Welcome message (page title);
4. Link to sent messages and new message;
5. Dark/light mode activator tab;
6. Login and logout link and info;

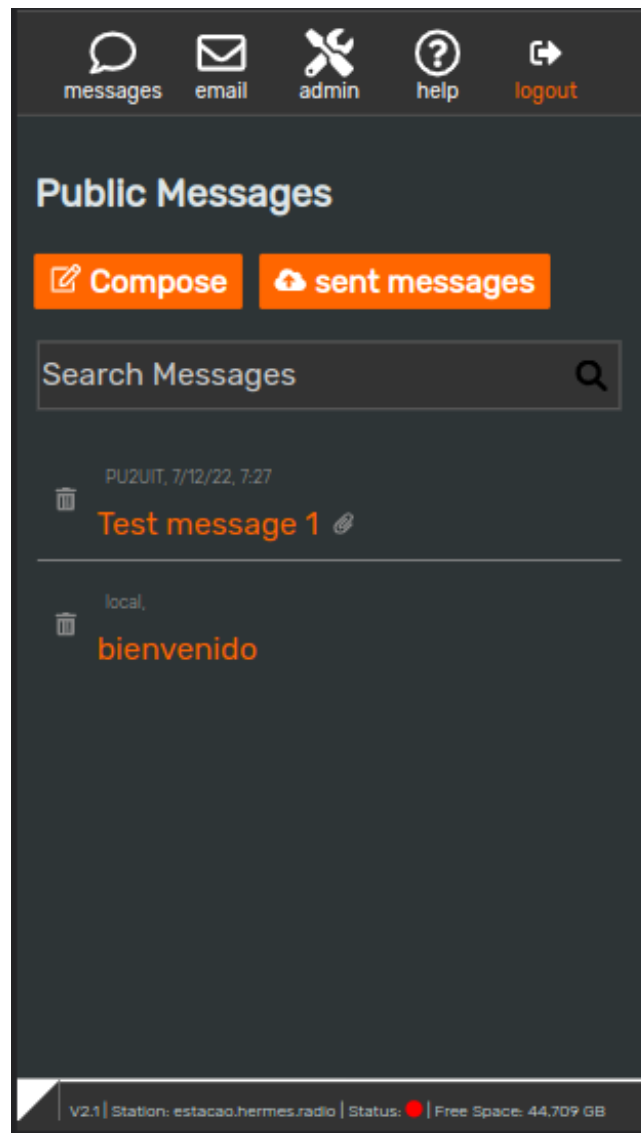


Figure 1.5: Hermes Dark Mode

The web interface allows admin users to manage e-mail accounts, to perform radio configurations (like setting frequency and SSB mode) and exchange direct messages between stations (BBS). The interface also contains its own administrative section, for defining and managing users and their privileges.

It is important to note that same user name created for the administrative interface will be used for the e-mail address assigned to that user. For example, user "amelia" in the local interface will have the email "amelia@ac2.hermes.radio". Note that the email address is composed of the user name @ the name of the station, which corresponds to its internet domain name, in this example, "ac2.hermes.radio". Each station domain name is written in the header on the Hermes web interface. If the domain begins with "ga" then it is a Gateway station.

1.2.1 Administrative Interface

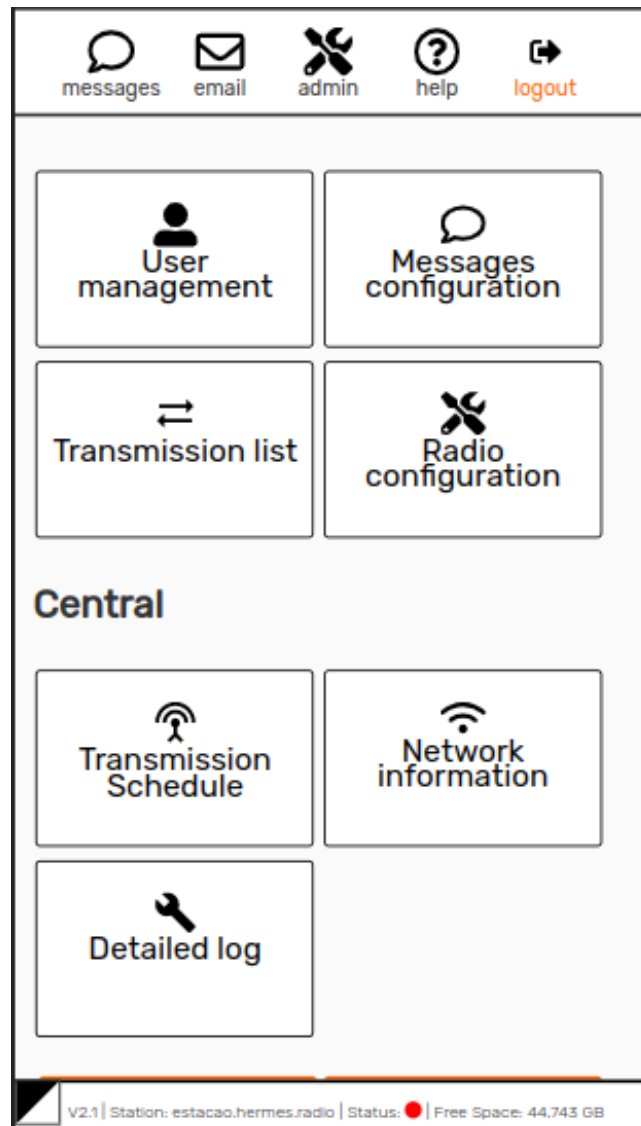


Figure 1.6: Admin interface

To access admin features, you need an admin password. While anyone can create a user account, only a system admin can create new administrator user and give admin power to other users. To login, click on the login icon on the top right of the web interface.

The default administrator login username is: "root" and the password is: "caduceu".

Inside the administration section, an admin user will find the following options: user management, messages administration, network information, stations, detailed log and radio configuration.

User management

Allows the creation of new users, updating data of registered users and to delete users of the system. Every username corresponds to an email account with the same name as in the example above (e.g. amelia@ac2.hermes.radio).

messages email admin help logout

User Management

Create User

email

write the username

@estacao.hermes.radio

full name

write your full name

phone number

phone number

password

password

Retype the password

is admin ☐

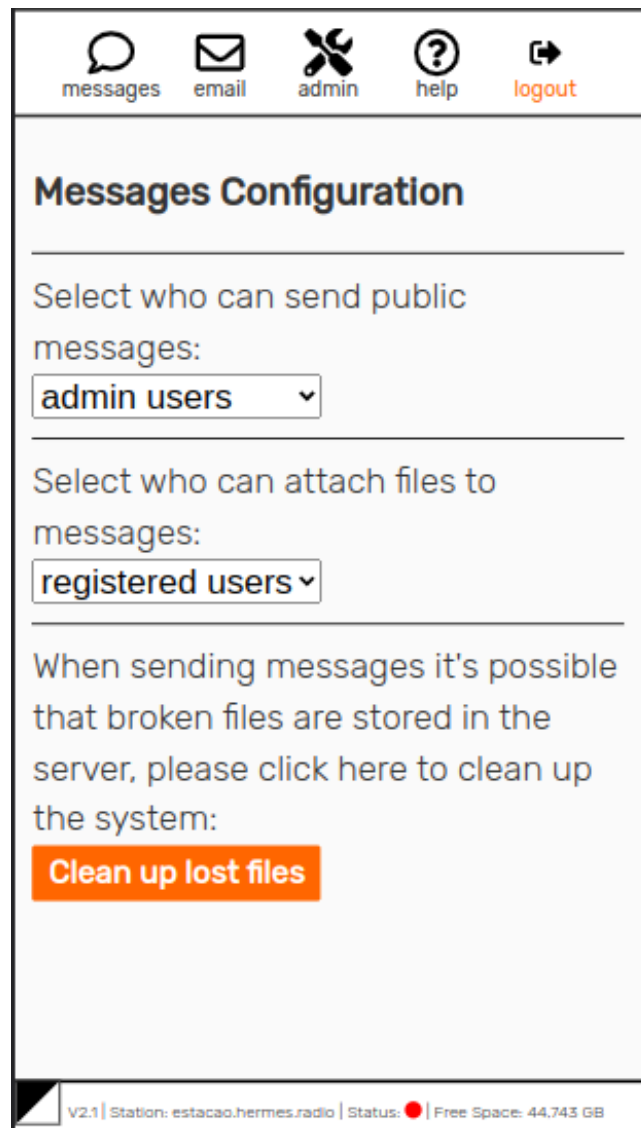
V2.1 | Station: estacao.hermes.radio | Status: ● | Free Space: 44,743 GB

Figure 1.7: Create user interface

Administrators can also give admin powers to regular users, by clicking on the "is admin" checkbox at the bottom of the Create User page.

Messages administration

From this page, an admin can determine who will be able to send public messages between stations and who can attach files to this messages: everyone; only registered users or only administrators. It is worth thinking this through with the users as attachments can slow the system down considerably.



The image shows a web interface for 'Messages Configuration'. At the top is a navigation bar with icons and labels for 'messages', 'email', 'admin', 'help', and 'logout'. The main content area has a title 'Messages Configuration' followed by two sections. The first section, 'Select who can send public messages:', has a dropdown menu currently showing 'admin users'. The second section, 'Select who can attach files to messages:', has a dropdown menu currently showing 'registered users'. Below these is a text block explaining that broken files might be stored on the server and providing a link to 'Clean up lost files', which is represented by an orange button. At the bottom of the interface is a footer bar containing the version 'V2.1', the station name 'estacao.hermes.radio', the status (indicated by a red dot), and the available free space '44,743 GB'.

Figure 1.8: Messages administration Interface

1.2.2 Transmission List

All data exchange in the HERMES system is done through UUCP. Being an asynchronous protocol, all the data is first queued before being transmitted. The elements of the UUCP queue are called "jobs", and each job in the HERMES system can be an e-mail, a public message, or special remote command execution message (for example, to inform of a new e-mail user creation). A system administrator can cancel a job before it is sent. Administrators can also erase the public messages for any reason, including the case when the equipment storage space is reaching its limit. The storage space available is shown in the web interface footer.

The queues of all Host stations are transmitted periodically when the Gateway station connects to each remote station based on whatever schedule has been configured. In an emergency, a system administrator can force message transmission and bypass the queue by clicking on the "force" link. This could interrupt receiving messages from other stations for a while and should be avoided unless absolutely necessary.

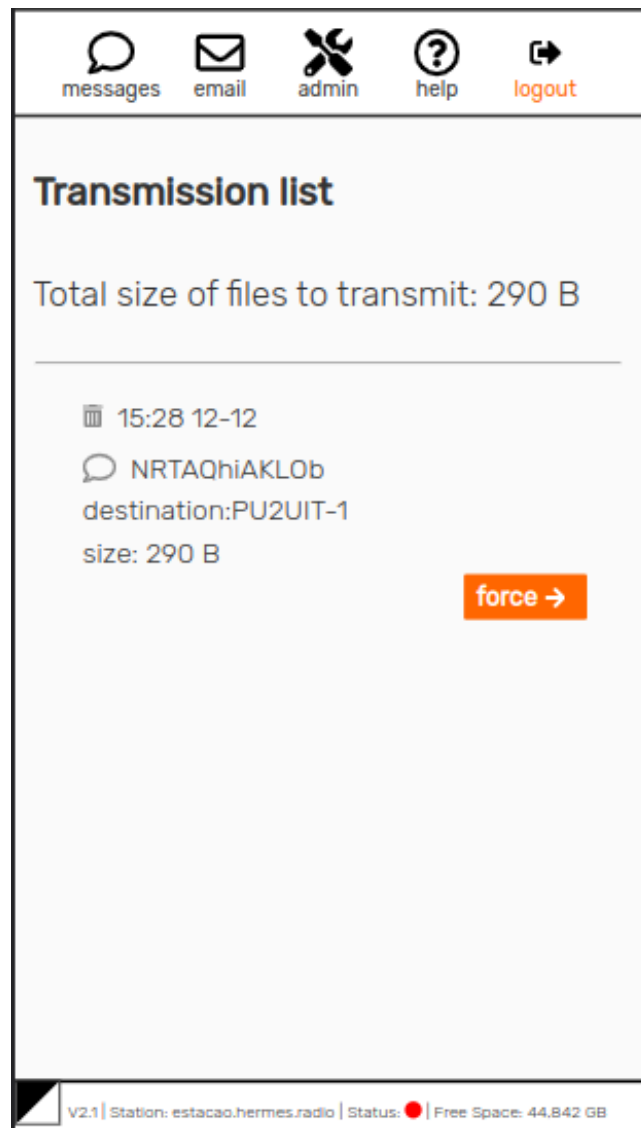


Figure 1.9: Transmission list interface

Radio configuration

Provides a direct interface to change various radio settings like frequency, transmission mode, restore factory configuration and it also allows one to view sensor readings about the HF antenna of the system.

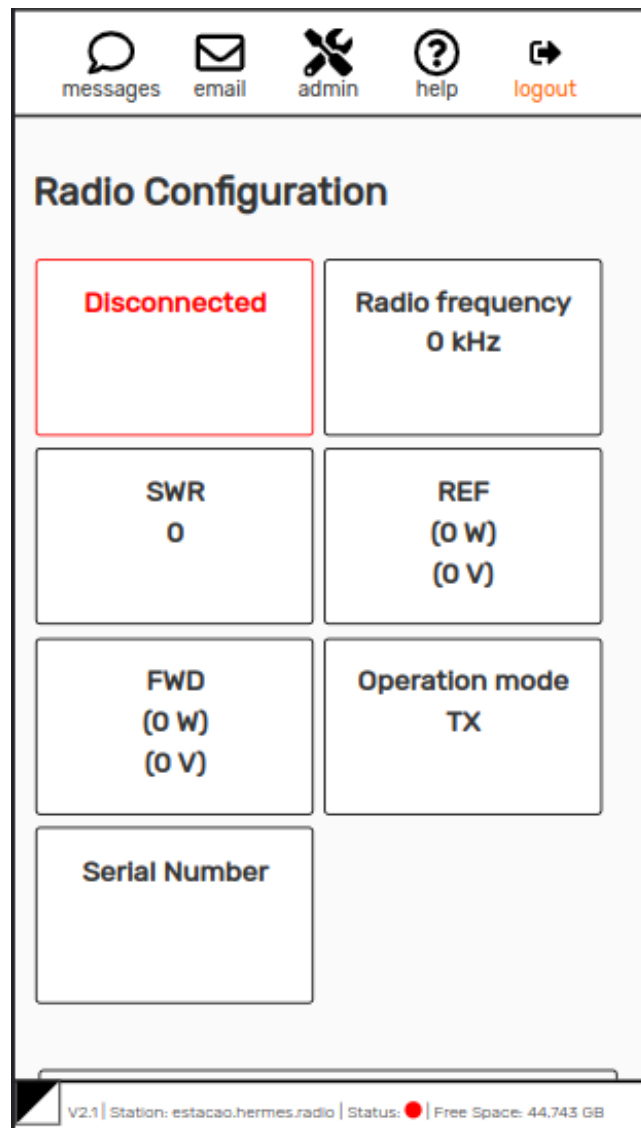


Figure 1.10: Radio configuration interface

messagesemailadminhelplogout

Radio frequency: 0 kHz

0

change frequency

Protection activation threshold level:

change threshold

Radio mode

USB ☒ LSB

PTT

☒

GPS Calibration

Start

V2.1 | Station: estacao.hermes.radio | Status: | Free Space: 44.743 GB

Figure 1.11: Radio configuration interface

Central Station

The central station is a special station that keeps connected to the internet. If you are in the central station, this option will be shown in the menu. From there is possible to create schedules for transmission to other stations, that are the time slots when the transmission will happen between all the stations connected to the network. It's possible also to enable or disable schedules or change their times from this menu.

messages

email

admin

help

logout

Transmission Schedule

create schedule

Current Schedules:

● teste

(12:00:00 - 15:00:00)

● Teste2

(18:00:00 - 19:00:00)

● Teste3

(17:00:00 - 17:30:00)

Schedule Title

Teste3

Start transmission hour:

V2.1 | Station: estacao.hermes.radio | Status: ● | Free Space: 44,743 GB

Figure 1.12: Transmission schedule

messages

email

admin

help

logout

(17:00:00 - 17:30:00)

Schedule Title

Teste3

Start transmission hour:

05:00 PM 🕒

Stop transmission hour:

05:30 PM 🕒

Select station:

× local ×

Enable ☐

change delete schedule

V2.1 | Station: estacao.hermes.radio | Status: 🟡 | Free Space: 44,743 GB

Figure 1.13: Transmission schedule

Network info

Displays some information about the system, such as network addresses, callsign, servername etc



Figure 1.14: Network information page

Stations

Provides a list of the available stations on the system.

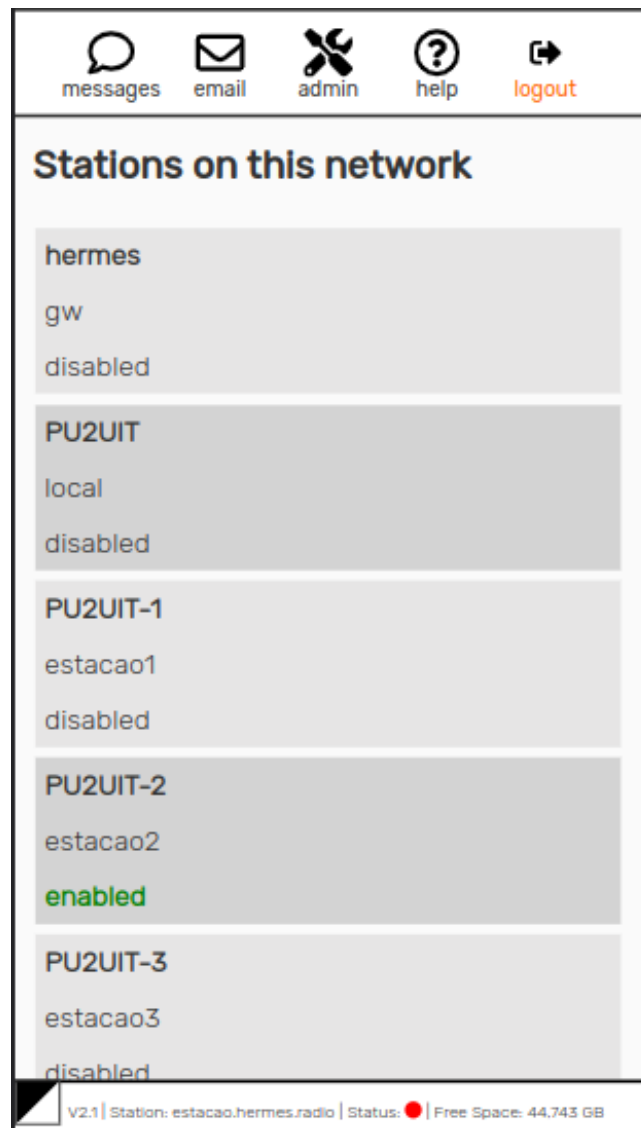


Figure 1.15: Stations interface

Detailed Log

Provides access to system logs, such as email logs and UUCP logs, that register every activity on the system.



Figure 1.16: Logs page on the web interface

1.2.3 Public Messages (BBS)

Direct messages can be sent between stations with support for cryptography and multimedia compression. Sent and received messages can be found in the Messages tab of the web interface. One Admin of the system can select who will be able to send public messages to other or tho it's own station.

How to write public messages

messages

email

admin

help

logout

send your public message to another station

Description:

public test

Message:

public message

'A' select station:

x estacao2

x estacao1

x

Attach File:

v2.1 | Station: estacao.hermes.radio | Status: | Free Space: 44,743 GB

Figure 1.17: Interface to compose messages

Figure 1.18: Interface to compose messages

By clicking on the compose () icon, it's possible to write a new message and to attach files such as images or audio files. The icon will only show up if it's allowed for the user to compose messages on the station. An admin can determine who can send public messages between stations and who can attach files to public messages by setting permissions in the admin tab.

Public messages can also be sent to your own station, which is an easy way to publicize news inside your own community.

Public messages can also be password protected, which means that only recipients that know the password will be able to read their content, while the message description will still be readable by everyone. Have in mind that once a password for the message (has to be at least 4 characters) is defined, there is no way to recover it, nor to change it.

On the message administration link in the admin section, a system administrator can change who can attach files to messages between stations: everyone with access to the network, only registered users, or only administrators.

Because the data throughput over HF is relatively low, file attachments are limited to 20KB. The system will accept inputs for image and sound up to 30 MB and for other files up to 2 MB, and will try to compress them to fit the maximum size of 20KB using state-of-the-art compression techniques.

Obviously, this can have an effect on both image and sound file quality. For other file formats, a simple compressor will be applied, and messages with attachment size greater than 20 KB will be cancelled.

1.2.4 Languages supported

The HERMES interface is also available in Portuguese and Spanish, and these versions can be accessed on the Language tab of the main menu.

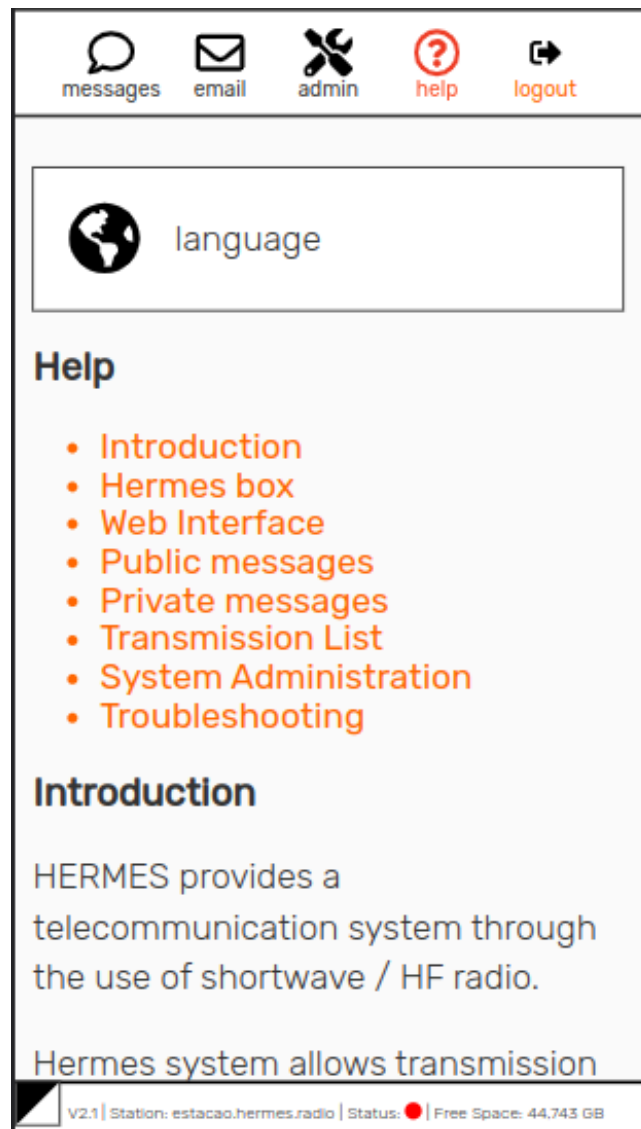


Figure 1.19: Page to access translations



Figure 1.20: Page to access translations

1.3 E-mail

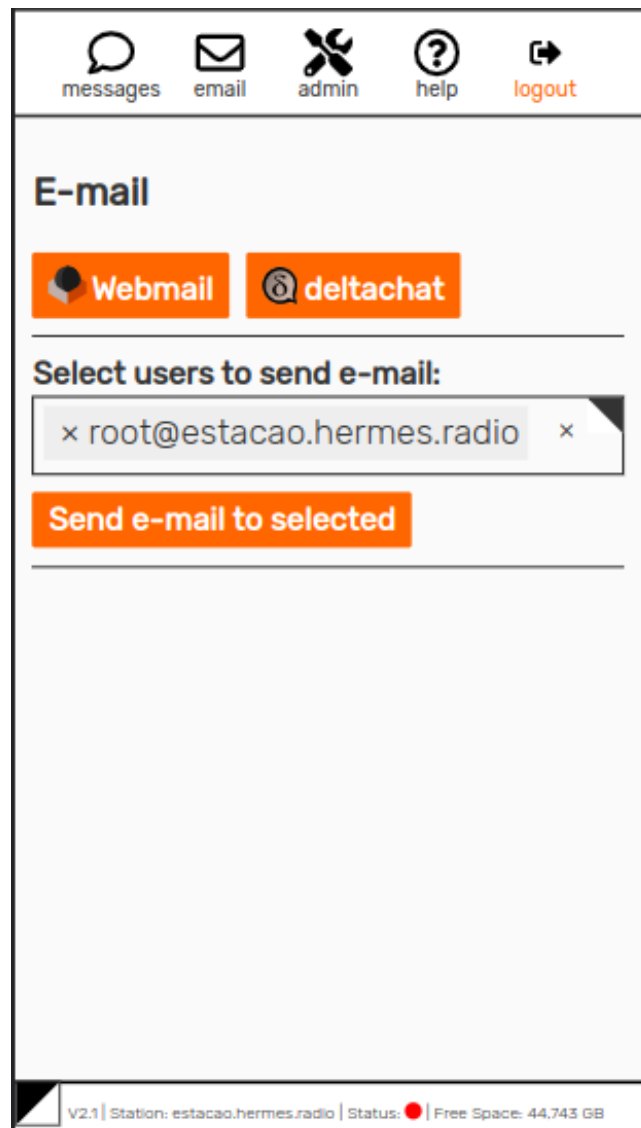


Figure 1.21: Page to access webmail interface

A key service provided by the HERMES is E-mail. The E-mail is communication protocol attributes to each e-mail bearer an address in the format "username@host". In the case of HERMES, the "username" part of the email is created by a HERMES user in the web interface, while the host (also called domain) is already set in the system, and typically has the format "community_id.hermes.radio". So a typical HERMES emails looks like, for example, "amelia@ac4.hermes.radio". The "username" is the same username as created in the user creation page in the HERMES administration interface.

HERMES email users can send and receive e-mail just like any other e-mail user. The only restriction which must be considered is that in order to avoid clogging the system with large messages, emails with large attachments will be cancelled by the system, with the appropriate cancellation message sent to the user.

While there are many e-mail clients, like Thunderbird and Outlook Express, the recommended e-mail client to be used with HERMES is DeltaChat. DeltaChat installers are available for download through the interface for Android, Windows, MacOS and Linux. A backup option is to use RoundCube webmail which is also accessible through the HERMES web interface.

1.3.1 DeltaChat

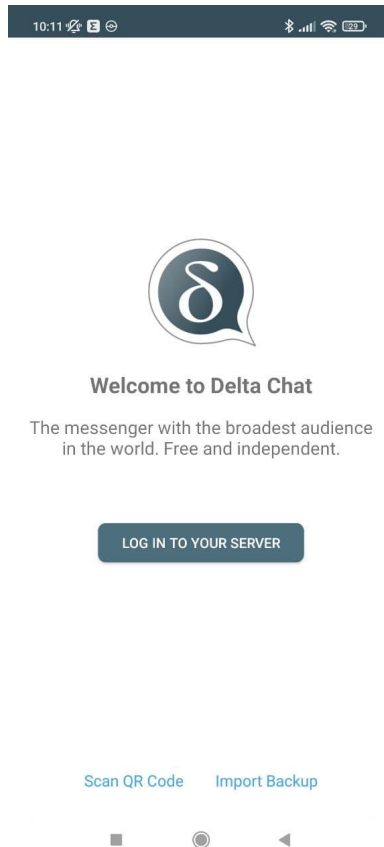


Figure 1.22: Deltachat intro (first screen)



Figure 1.23: Deltachat chat room example

With the DeltaChat application, you can use HERMES email to carry out personal communications for exchanging messages. This app works on most common smartphones and feels like a common messaging application like Telegram, WhatsApp or Signal. Keep in mind that due to the message transmission schedule, messages may take a while to arrive, depending on the amount of messages in the queue and the transmission window length.

Installation

DeltaChat is available in most common app stores and repositories. As HERMES is designed for places with low or no connectivity, it is possible to download it through the HERMES web interface, selecting one of the packages provided according to your device operational system. HERMES provides installation files for mobile or computer systems such as Android, GNU/Linux, Windows and MacOSX. The installer files can be accessed here if you're reading this using a HERMES system network: [Android](#), [Windows](#), [Debian](#) and [Mac OS](#)

Configuration

The HERMES system includes a compression system suitable for multimedia messages like images or audio being sent over HF. End-to-end encryption should be disabled in order to allow the server-based image and audio compression to work properly, otherwise image and audio exchange will not be possible.

The steps to find this feature in Deltachat are: Burger Menu (☰) -> Settings -> Advanced -> Autocrypt, turn off: Prefer End-To-End Encryption) as shown in Figure 1.24.

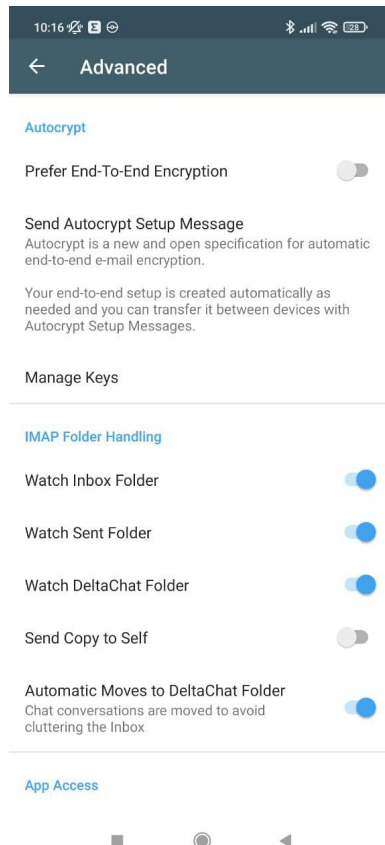


Figure 1.24: Deltachat Advanced settings

Deltachat, by default, tags its emails and show only the "known" messages (emails) that was sent

by another Deltachat application. In order to be able to interact with e-mails from any e-mail client software, enable the option "show all emails". The steps to find this feature in Deltachat are: Burger Menu (☰) -> Settings -> Chats and Media -> Show Classic E-Mails -> All. As shown in Figure 1.25

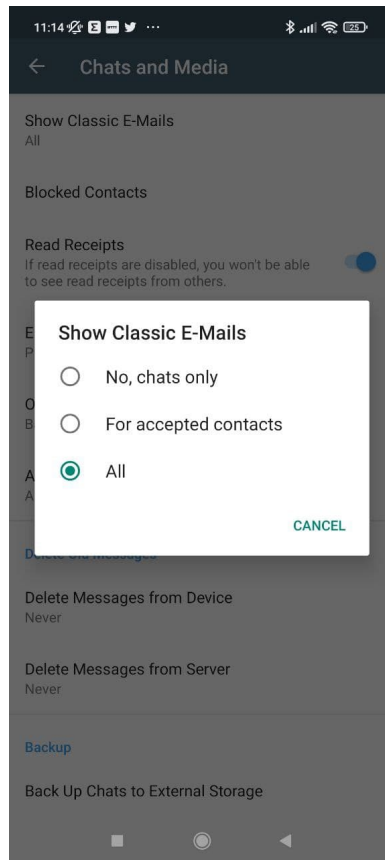


Figure 1.25: Deltachat Setup all emails

Usage

In order to configure DeltaChat, first an e-mail account should be created, as described in section 1.2.1. In order to login to an e-mail account, only the e-mail address and password fields should be filled, as shown in Figure 1.26. The user and password will be the same as those created when first creating a login on HERMES and for use with the RoundCube webmail.

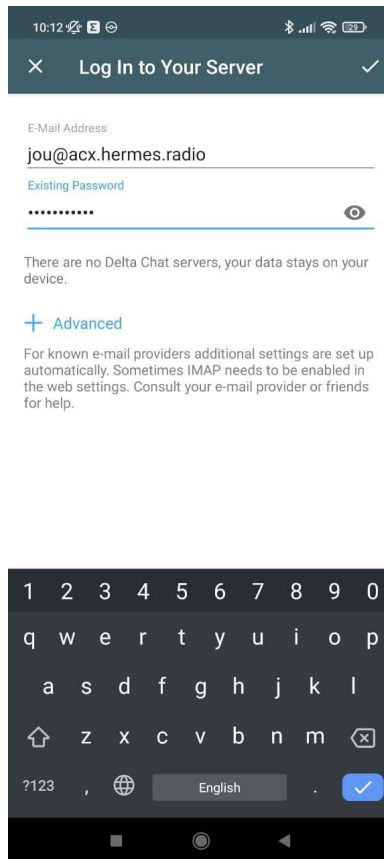


Figure 1.26: Deltachat Login example

1.3.2 Other email clients

Other email clients can also be used. We can't cover all cases, but the HERMES system uses default services like IMAP to synchronize email folders and SMTP to send the messages. Specific technical information about the ports can be found in the appendix: [1.5.2](#)

1.4 Troubleshooting

The HERMES web interface indicates when something is wrong with the system. When a red dot appears in the footer bar (figure 1.27), it becomes interactive, and when clicked, it lists the names of the services that are experiencing a problem.



Figure 1.27: Red dot on the web interface can be clicked to check what part of the system may be broken

In case of problems with the antenna, a red led in the antenna (ANT) position will light up on the HERMES box (not the web interface). In this case, check the correct position of the power supply wires and check the position and integrity of the antenna, RF connectors, coaxial cable, and the transmitter. After checking that, it will be necessary to rest the antenna protection on the radio config section on the web interface for the radio transmit again.

If the messages in the transmission list are not being transmitted at the allotted time, check that the

radio settings have not been changed by going to the Admin tab. If that is not the case, it's possible that the other stations are offline.

1.5 Appendix - technical information

Maintainers note: this user manual focuses on station operation and the web interface. For a detailed HERMES system architecture and software stack description (e.g. `uucpd/uuport`, `sbitx_controller`, modem/TNC integration, and systemd service orchestration), see the document “**HERMES Software Description**” (`software_en.pdf`) in the HERMES documentation repository.

1.5.1 Network Information

Each HERMES system provides a WiFi access point. The configuration is:

- WiFi Network Name: `HERMES`
- WiFi Password: `amazonia`

The IP of the HERMES station (in the wireless interface) is **10.0.0.1** and the ethernet connection will connect as a dhcp client to an already existing network.

1.5.2 Network Services

E-mail services

The e-mail settings are the following:

- Server address: `hermes` (IP `10.0.0.1`)
- SMTP (Simple Mail Transfer Protocol): port 25
- SMTPS (Secure Simple Mail Transfer Protocol): port 465, with SSL support (used for DeltaChat, the certificates are valid until 2031)
- IMAP (Internet Message Access Protocol): port 143
- Webmail URL accessible: `http://hermen/mail` or `http://10.0.0.1/mail`

These settings should not be needed by most users as DeltaChat automatically identifies all the settings by entering just the e-mail address and password at the login prompt.

Web services

The HERMES web interface runs on port 80 on http, and provides these capabilities:

- BBS P2P public messaging capability (over UUCP);
- E-mail user administration;
- UUCP queue management;
- HF radio transceiver management tools;

- Customized permissions for users send multimedia messages;
- Page for radio service tasks (like test tone generation and PTT control);
- A network information page (for cases in which the HERMES system connects to a larger IP network);
- DeltaChat client downloads for Android, MacOS, Windows and Linux.

1.5.3 Other network Services

- SSH Secure Shell - for special system administration: port 22
user: "hermes", password: hermes
user: "root" password: "caduceu"
- VPN - Virtual Private Network client ready for remote access
- ISPCONFIG admin web interface
port 8080 / credentials: "admin" "caduceu"
- mariaDB, the Database server for storing messages and users on station-api and ISPconfig
- E-mail services with transports connected to UUCP
Postfix, dovecot, spamassin, postgrey, amavis, clamav (hold) and ISPconfig as a manager
- iwatch: handles inbox HMP folder and triggers UUCP spool compression
- hostapd: sets the wireless interface into Access Point mode
- dnsmasq: provides a domain name server and aliases
- uucp: accessible via network with credentials user/pass
- VNC: Virtual X session environment with VARA monitor port: 5901 (Eg. in the local wifi network: vncviewer 10.0.0.1:5901) user: "hermes", password: "hermes"

1.6 Additional information

The main system runs Debian GNU/Linux release Bullseye and we try to follow their guidelines.

1.6.1 Password Cheat Sheet

Description	User	Password
Web Interface	root	caduceu
WiFi Network	hermes	amazonia

1.6.2 Field trials

HERMES system prototypes were tested on a test bed with stations installed in three cities: Brasília/DF, Belo Horizonte/MG and Hortolândia/SP. Most of the tests happened between Brasília and Belo Horizonte, and Belo Horizonte and Hortolândia. The straight line distance between Brasília and Belo

Horizonte is 620 km, while Belo Horizonte and Hortolândia straight line distance is 470 km. All the stations are equipped with simple dipole installed as inverted V, tuned to the 40m amateur radio band.

In our internal tests, between Belo Horizonte and Hortolândia, the modem reached more than 1000 bps in an average propagation condition (0db of SNR in the receiver). A 10Kb message, which is the typical size of an email with a picture takes about 4 minutes to be exchanged. In poor propagation conditions, this time can go up to 10 minutes.

The adaptive modem starts the communication with slower speeds, but if propagation is good, it gears up and automatically and increases the speed, and on the other hand, if propagation deteriorates, the modem reduces the speed, increasing the signal robustness.

1.6.3 Source Code

Source code is available inside the folders /home/hermen/install with the latest git versions before the deployment and is also available online at:

- Web Front-end Interface, written using the Angular framework: <https://github.com/Rhizomatica/hermes-gui>;
- Web Back-end Interface, written in PHP: <https://github.com/Rhizomatica/hermes-api>;
- HF transceiver description and schematics: <https://github.com/DigitalHERMen/rhizo-transceiver>;
- HF transceiver firmware and userland tools, written in C: <https://github.com/Rhizomatica/hermes-net>;
- UUCP-to-modem bridge (uucpd + uuport) for VARA-style TNC integration, written in C: <https://github.com/Rhizomatica/hermes-net/tree/main/uucpd>

1.7 Licensing

All the project's source code is licensed under the GPL version 3 or any greater version, unless stated differently in the repository.

1.7.1 GNU General Public License Version 3

High-Frequency Emergency and Rural Multimedia Exchange System (HERMES).

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Chapter 2

sBitx installation guide

2.1 Introduction

This guide goes through the steps needed for HERMES system installation.

2.2 Power supply and consumption

In order to connect the radio to the power source, a power cable is provided. One side of the cable comes with a XT60 connector (Figure 2.2) and connects to the radio, and the other end is bare ended, and should be connected to the positive (Red) and negative (Black) of a 12V battery system, or power supply. Power consumption is not more than 1.5 A in receive mode, and 9 A in transmit mode.

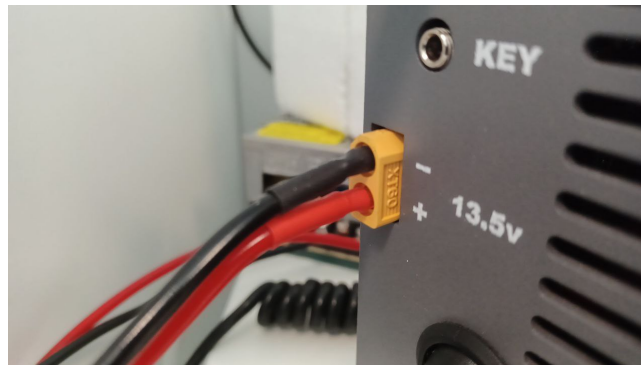


Figure 2.1: The radio-side of the power cables where the XT60 connector is located.

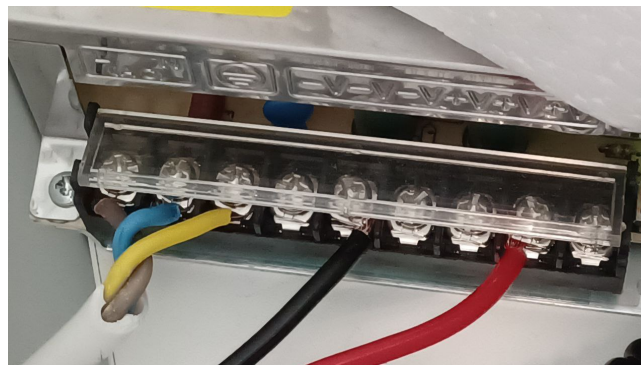


Figure 2.2: Red (+) and black (-) cables connected to appropriate power supply terminals.

2.3 Initial setup to allow boot from USB pendrive

This step is needed in order to enable the HERMES installer to run from a USB pendrive.

First connect a USB keyboard to the radio, and then connect the radio to the network (RJ45) cable with appropriate Internet connection. Then click in the Terminal icon shown in Figure 2.3.

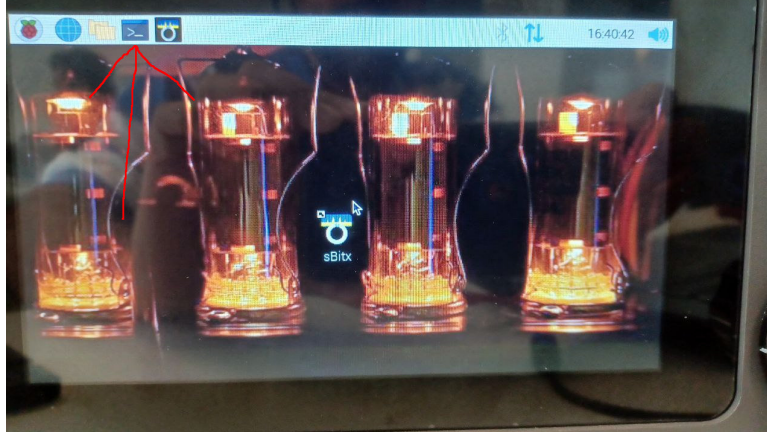


Figure 2.3: Screen of the radio and the “Terminal” icon pointed by a red arrow.

After opening the terminal window, type:

- `sudo su`
- `apt-get update`
- `apt-get install rpi-eeeprom`
- `raspi-config`

At this point the screen shown in Figure 2.4 should appear.

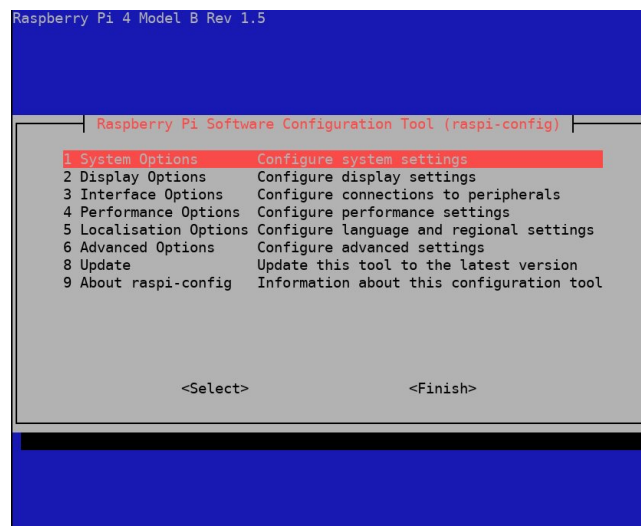


Figure 2.4: Configuration software screen (raspi-config tool).

Choose the options of the menu in the following order using the keyboard arrow and Tab keys (as shows in Figures):

- Step 1: Option 6 (Advanced Options)

- Step 2: Option A6 (Boot Order)
- Step 3: Option B2 (USB Boot)
- Step 4: Finish

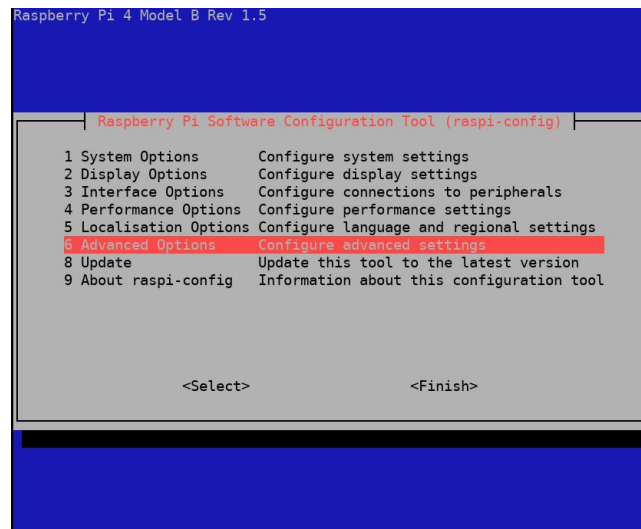


Figure 2.5: Menu option to select (step 1).

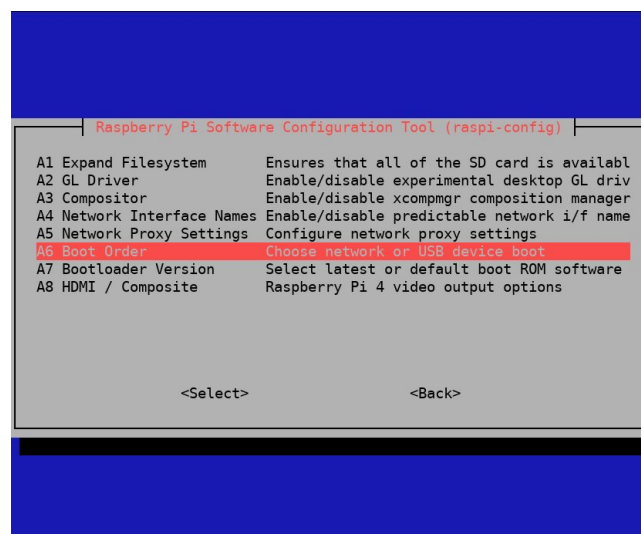


Figure 2.6: Menu option to select (step 2).

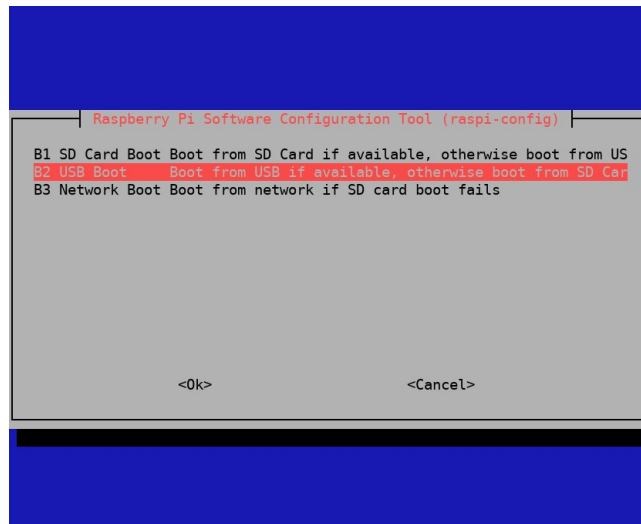


Figure 2.7: Menu option to select (step 3).

Then just accept the option to reboot the radio. Now the radio is ready to run the HERMES installer. If needed, in order to turn off the radio, type the following in the terminal window (just after raspi-config):

- halt

And after some seconds, when the screen goes black, turn off the radio (using the power key below the power socket).

2.4 HERMES USB installer preparation

A USB pendrive of at least 8 GB is needed. A computer is needed to prepare the USB pendrive.

The installer should be downloaded from: <https://aco-connexion.org/downloads/installer-image.img.zip>.

After the download is finished, the file should be “unzipped”, as shown in Figures 2.8, 2.9 and 2.10. The extracted file should have the name: “installer-image.img”.

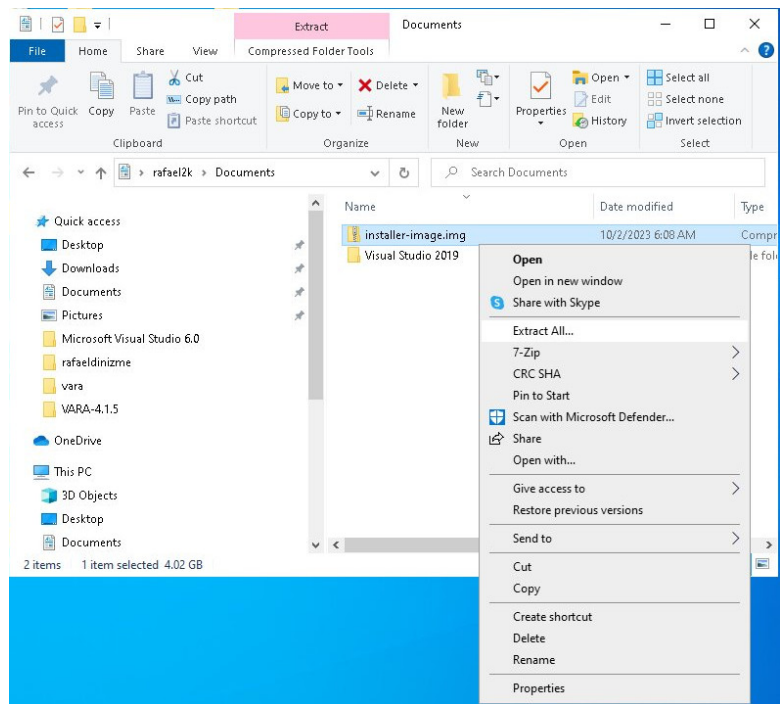


Figure 2.8: Unzipping the installer file.

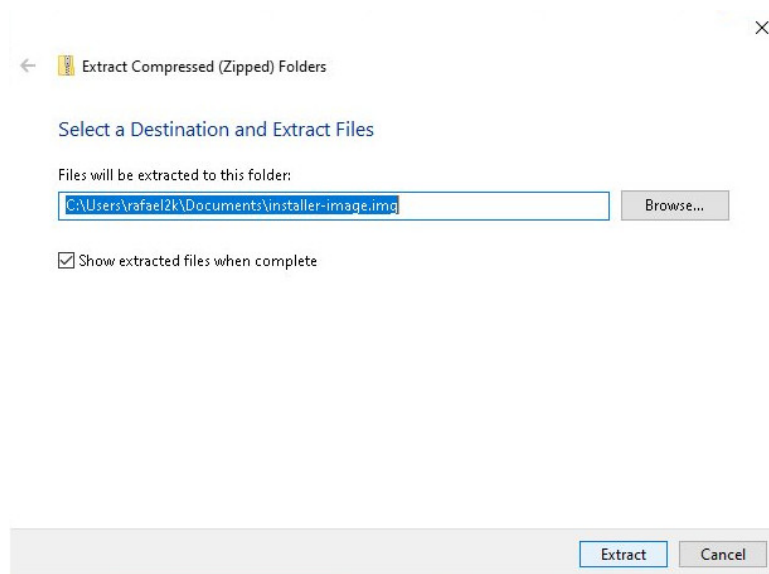


Figure 2.9: Setting the installer file name.

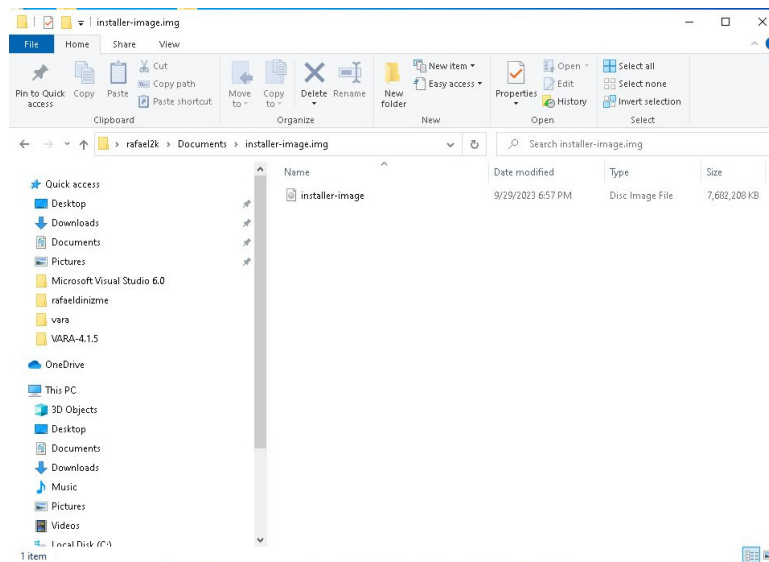


Figure 2.10: Installer file after unzip.

Then, for copying the installer to the USB pendrive, use the following software (or other raw image writer or flasher), called Etcher (supported on Windows, MacOS and Linux), available in <https://etcher.balena.io/>. The download is available on this website, as shown in Figure 2.11 (eg: for Windows choose the first option).

DOWNLOAD

Download Etcher

ASSET	OS	ARCH	
ETCHER FOR WINDOWS (X86 X64) (INSTALLER)	WINDOWS	X86 X64	Download
ETCHER FOR WINDOWS (X86 X64) (PORTABLE)	WINDOWS	X86 X64	Download
ETCHER FOR WINDOWS (LEGACY 32 BIT) (X86 X64) (PORTABLE)	WINDOWS	X86 X64	Download
ETCHER FOR MACOS	MACOS	X64	Download
ETCHER FOR LINUX X64 (64-BIT) (APPIMAGE)	LINUX	X64	Download
ETCHER FOR LINUX (LEGACY 32 BIT) (APPIMAGE)	LINUX	X86	Download

Figure 2.11: Download options for Etcher software.

Run the software Etcher, choose the option “Flash from file” and select the file “installer-image.img”, as shown in Figures 2.12, 2.13 and 2.14.

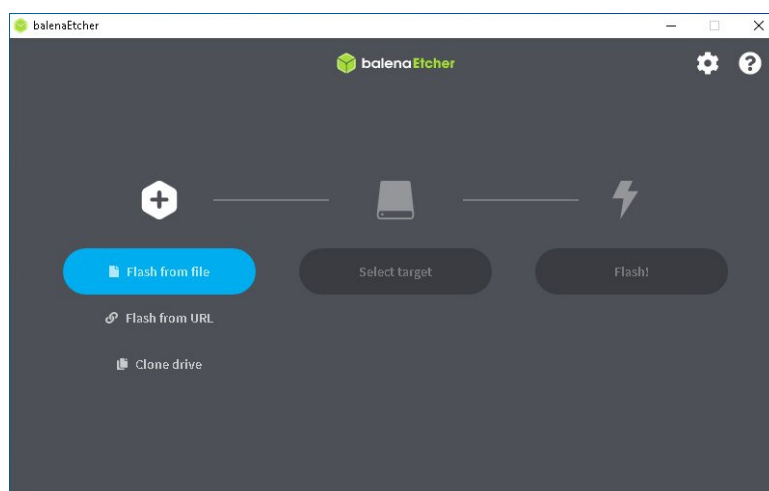


Figure 2.12: First screen of Etcher, “Flash from file” selected.

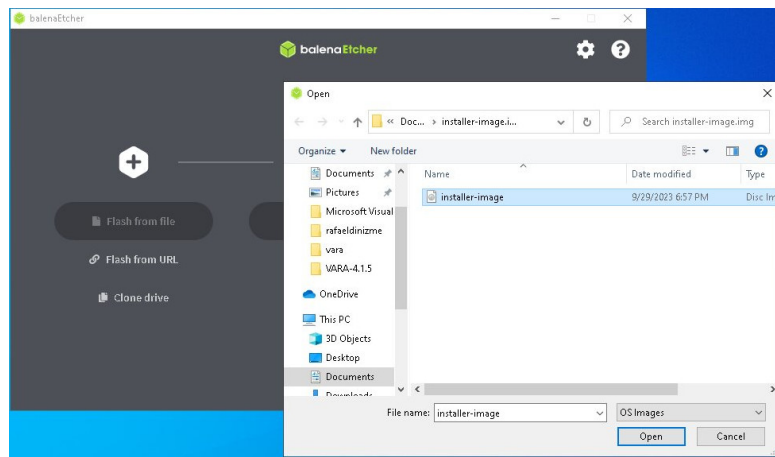


Figure 2.13: Choose the file “installer-image”.

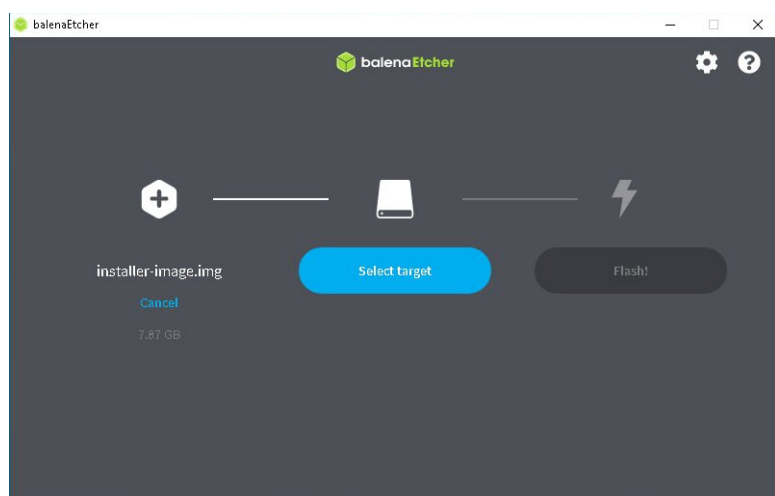


Figure 2.14: Select the USB pendrive to be used as installer.

After the copy is finished, insert the USB pendrive with the HERMES installer in the radio, as shown in Figure 2.15. Make sure the radio is turned OFF before inserting the USB pendrive.



Figure 2.15: Radio with USB pendrive connected, together with keyboard and network cable.

2.5 HERMES installation

The HERMES installation procedure will set up the sBitx radio with HERMES system, and perform the station-specific setup.

The installation is done in two steps. The first one with the pendrive inserted in the radio, and the second without the pendrive.

After preparing the USB pendrive and plugging it to the radio, turn on the radio.

When asked if you want to proceed with the installation, answer “y” and press ENTER, as shown in Figure 2.16.

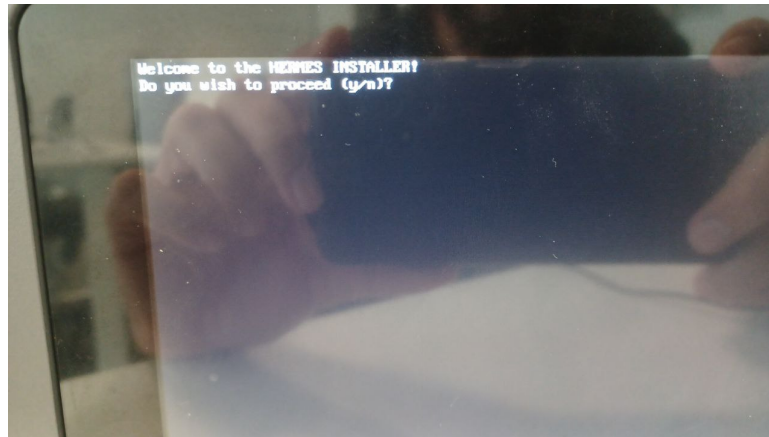


Figure 2.16: HERMES interactive installer starting.

When asked if you want to write HERMES (copy the system), answer “y” and press ENTER, as shown in Figure 2.17.

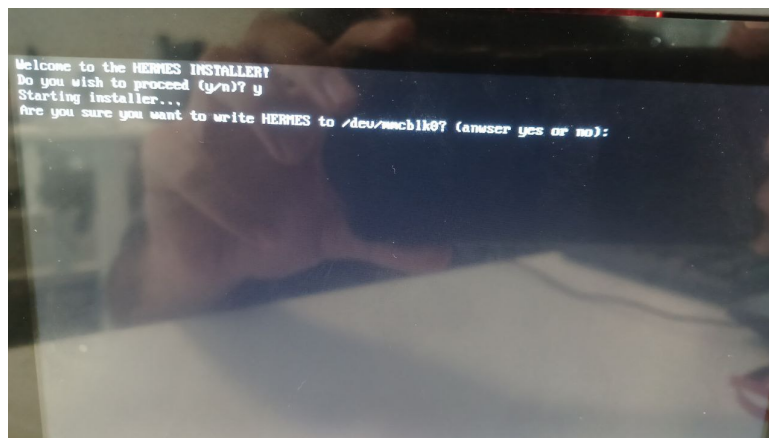


Figure 2.17: HERMES installer question to start writing to internal radio memory.

Then the installer will copy the HERMES system (Figure 2.18) to the internal radio SD card, which will take some time, most likely more than one hour.

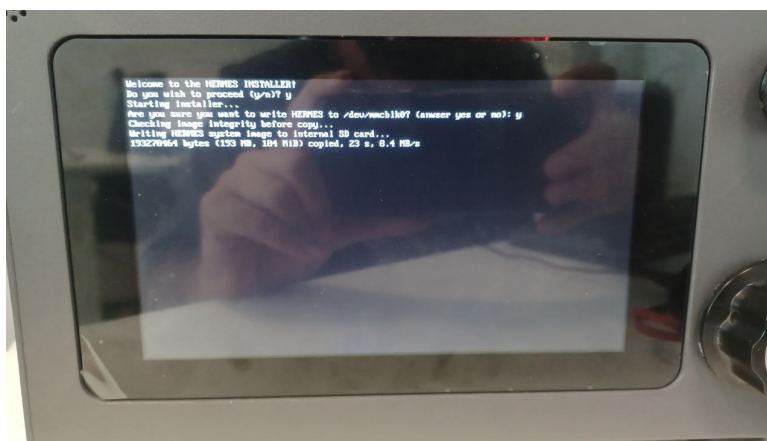


Figure 2.18: HERMES installer in process of copying data to internal radio memory.

After the copying is done (shown in Figure 2.19), reboot the radio (just press any key) and REMOVE the pendrive from the radio for the second step of the installation. Also make sure the network cable (Internet) is connected to the radio.

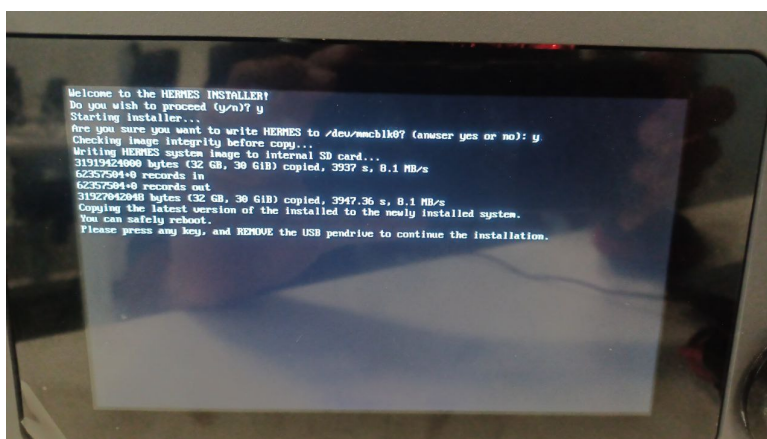


Figure 2.19: HERMES installer finished copy of the system to internal radio memory.

After the radio restarts, you will be asked about the station name to use, as shown in Figure 2.20 and 2.21. Choose one of station names below, by typing the full station name in the installed prompt.

- bakouma.aco-connexion.org
- bangassouhub.aco-connexion.org
- briaahub.aco-connexion.org
- goboudou.aco-connexion.org
- irabanda.aco-connexion.org

The station “bangassouhub.aco-connexion.org” has the gateway station setup. The gateway station should be deployed in a place with Internet connection.

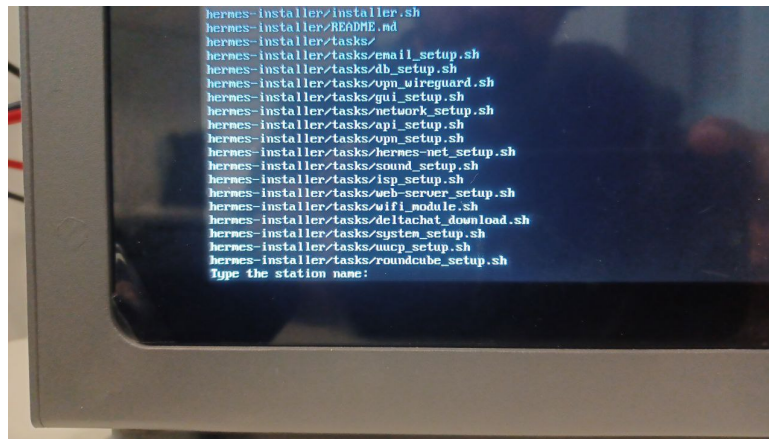


Figure 2.20: HERMES installer station name prompt.

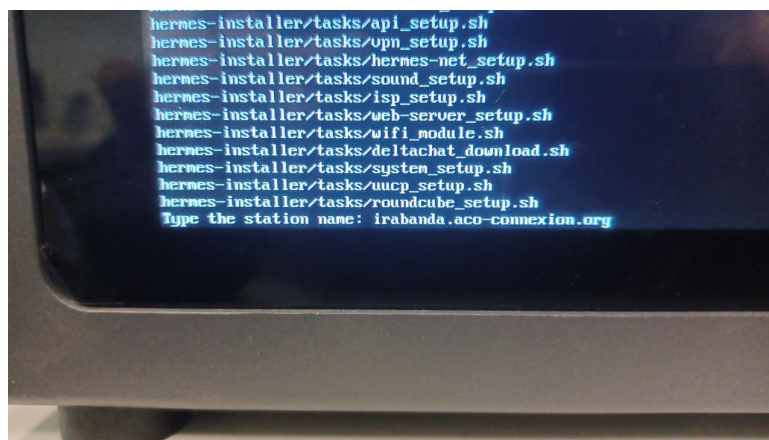


Figure 2.21: Station name field filled with “irabanda.aco-connexion.org”.

After the setup is finished, you should see the following message “Press any key to REBOOT the radio...”, as shown in Figure 2.22. Press any key. The installation process is finished.

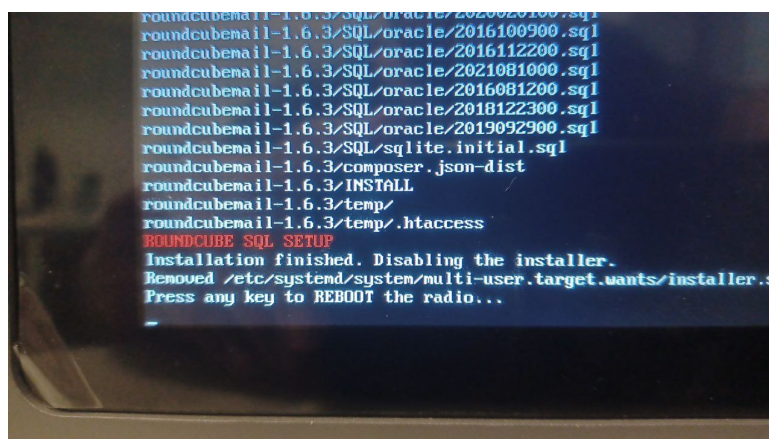


Figure 2.22: Message that marks the end of installation.

The radio will restart, and the HERMES interface will show up, as shown in Figure 2.23.

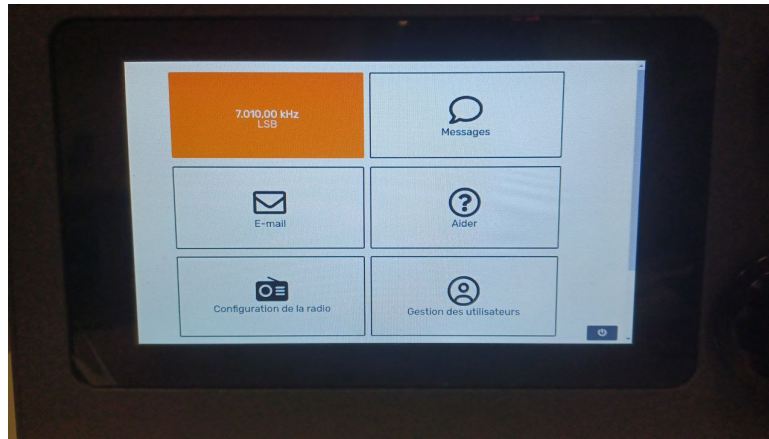


Figure 2.23: HERMES system screen.

2.6 WiFi Setup

Default radio's wireless information is:

- Network Name (SSID): hermes
- password: amazonia

In case the system connecting to the wifi of the radio shows a question saying that there is no internet, questioning if you want to stay connected anyway, select the option to continue connected anyway “always” (in case there is also an option to take this action “only once”).

The Wifi settings can be changed through the HERMES interface, as shown in Figures 2.24 and 2.25. Choose a network name only with lower case letters and numbers. The password needs to be at least 8 characters, and it is not allowed the use of special characters or spaces - use just lower case letter and numbers. Wifi channel also needs to be chosen, being the allowed channel numbers the range from 1 to 11. Any channel can be chosen. In case two or more HERMES radio are in the same room, choose a different channel and network name for each radio.

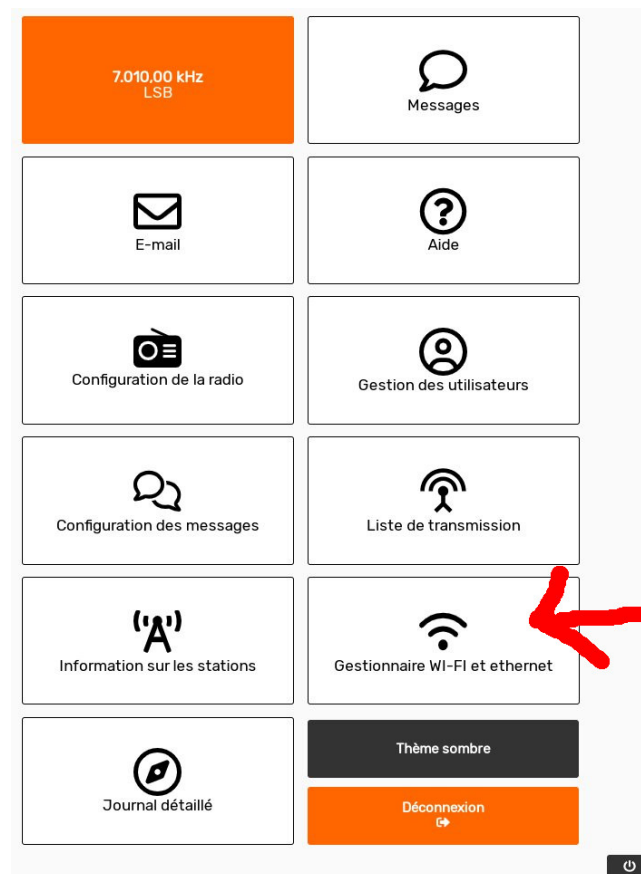


Figure 2.24: WiFi administrative interface icon pointed by red arrow.

< /home /wifi-manage

Gestionnaire Wi-Fi et ethernet

Nom du réseau SSID

Canal

Mot de passe

Confirmer le mot de passe

sauvegarder les modifications

Figure 2.25: WiFi administrative interface.

2.7 Radio Frequency (RF) connections assembly

For proper testing, the radio needs to be connected, ideally, to a power meter (also called wattmeter), which should be connected to a dummy load, as shown in Figures 2.26 and 2.27.



Figure 2.26: Back of the power meter with the cables attached.



Figure 2.27: Both the dummy load (left) and the power meter (right).

The radio (sBitx) has a BNC female connector, and needs the appropriate cables and adapters to be connected to the Wattmeter and dummy load (eg. by using BNC Male to UHF/SO239 Female). The wattmeter measures the forward and reflected power of the radio, and the dummy load simulates an antenna without radiating the signal.

The radio needs to be connected (see Figure 2.29) to the “TX” or “Transmitter” (see Figure 2.28) port of the power meter, while the dummy load should be connected to the “ANT” or “Antenna” port.



Figure 2.28: Back of the power meter in which the ANT and TX writings can be seen.



Figure 2.29: The power meter (left) and radio with the connected cable (top right).

Make sure the cables are properly connected and the front of the power meter (Figure 2.30) is in a visible position.



Figure 2.30: Front view of a power meter

2.8 RF test

After all the RF connections ready, a transmit test needs to be made. First, set a frequency in the band you will operate most the radio (Eg. 6200 kHz). Then two tests should be carried:

- Voice test with the microphone, for microphone and its button operation check
- Single tone test, for establishing the power level of the radio

In order to log-in to the HERMES User Interface (UI), a default password is set (that can be changed), and it is:

- user name: root
- password: caduceu

To enter the voice screen in HERMES UI, follow the Figures 2.31 and 2.32.



Figure 2.31: HERMES UI with a finger pointing to the voice functionality icon.



Figure 2.32: HERMES UI in the voice screen.

The expected result of the voice test, when talking with normal voice level, should be the wattmeter having a oscilating power level, ranging from 0 to 40W.

In order to carry the single tone test, enter the radio configuration screen, as shown in Figure 2.33. The single tone functionality is enabled using the “PTT” button in the radio configuration page, as shown in Figure 2.34.

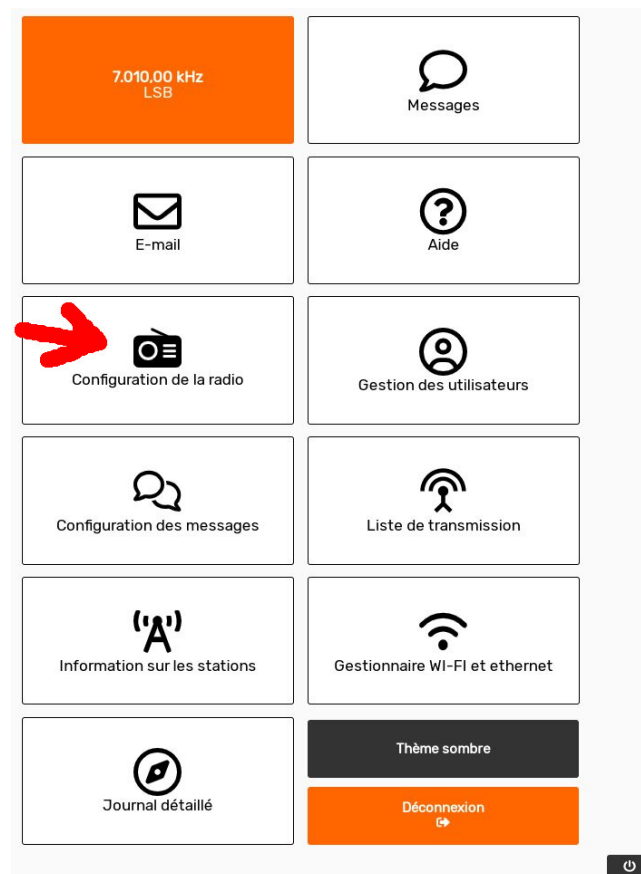


Figure 2.33: HERMES UI with an arrow pointing to the radio configuration icon.

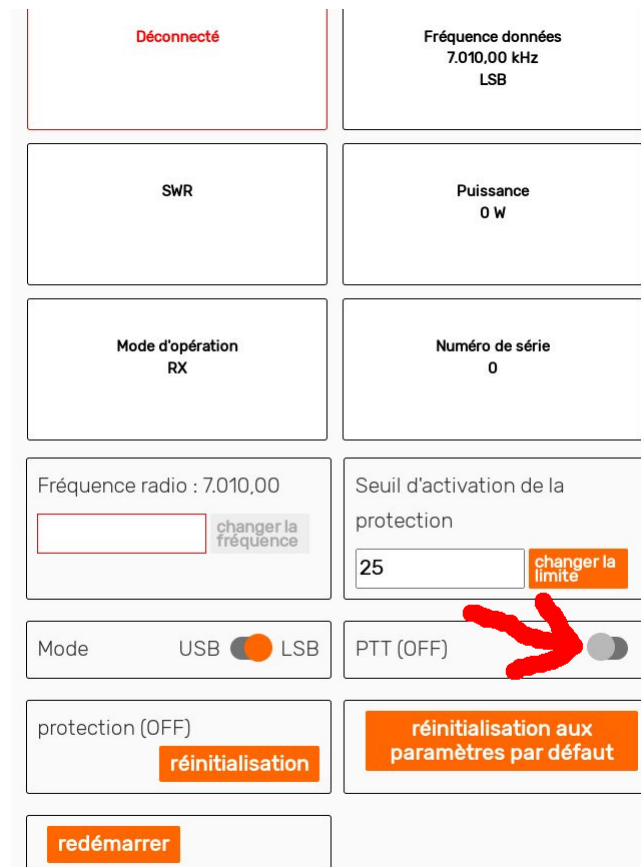


Figure 2.34: HERMES UI in the radio configuration page, an arrow points to the “PTT” button.

The expected result of the single tone test, should be a power reading of approximately 20 W, as shown in Figures 2.35 and 2.36.

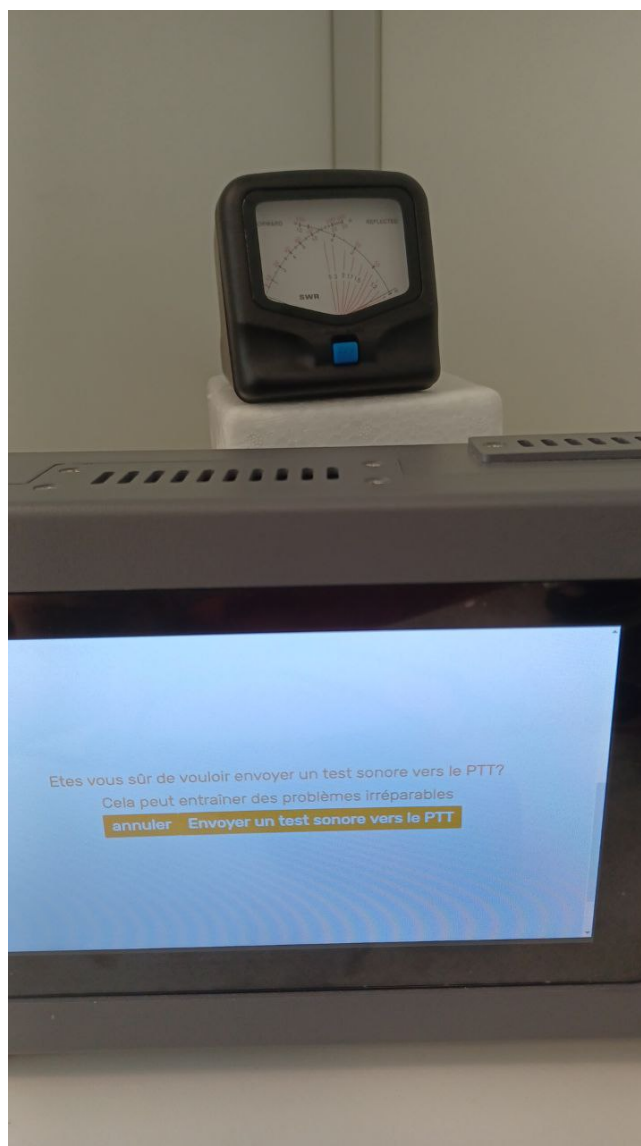


Figure 2.35: Confirmation question shown after clicking to enable the PTT. No power.

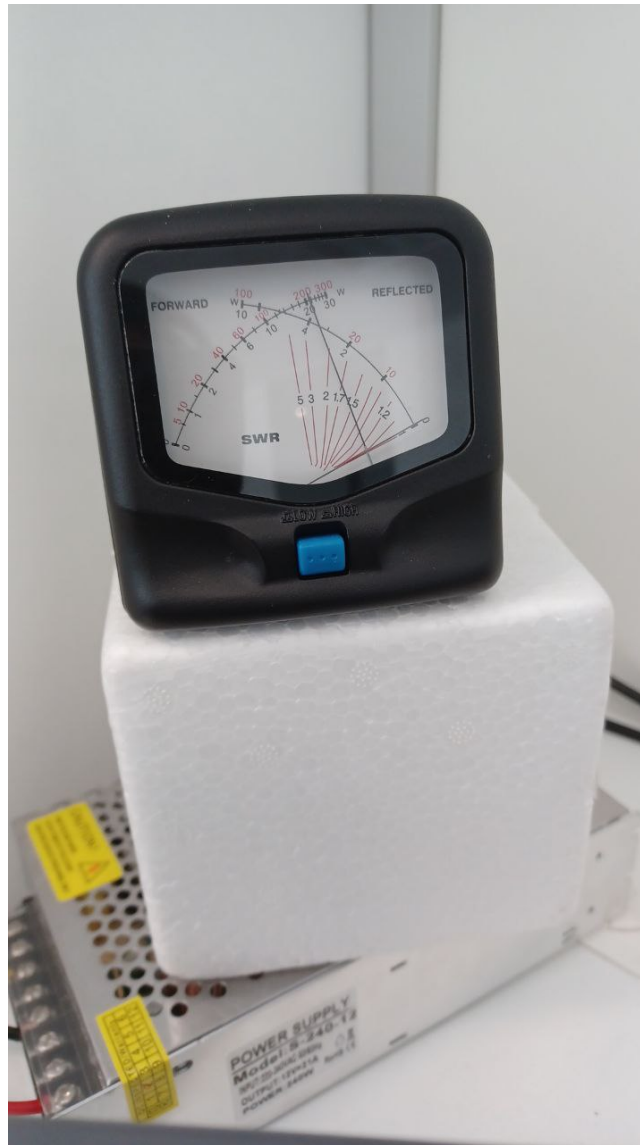


Figure 2.36: Wattmeter showing 20W of power.

2.9 E-mail setup and test

In this section the process of creating one or more users in the HERMES system is address. Also the process of connecting a device (eg. tablet or cellphone) to the radio over WiFi and using the software DeltaChat (an email client) for messages exchange.

In the HERMES interface, create one or more users, to be used for the email access. Then connect the companion device to the radio WiFi network, and install the software DeltaChat (an email client). Follow the steps shown in Figures 2.37 and 2.38 in order to create users in the HERMES UI.

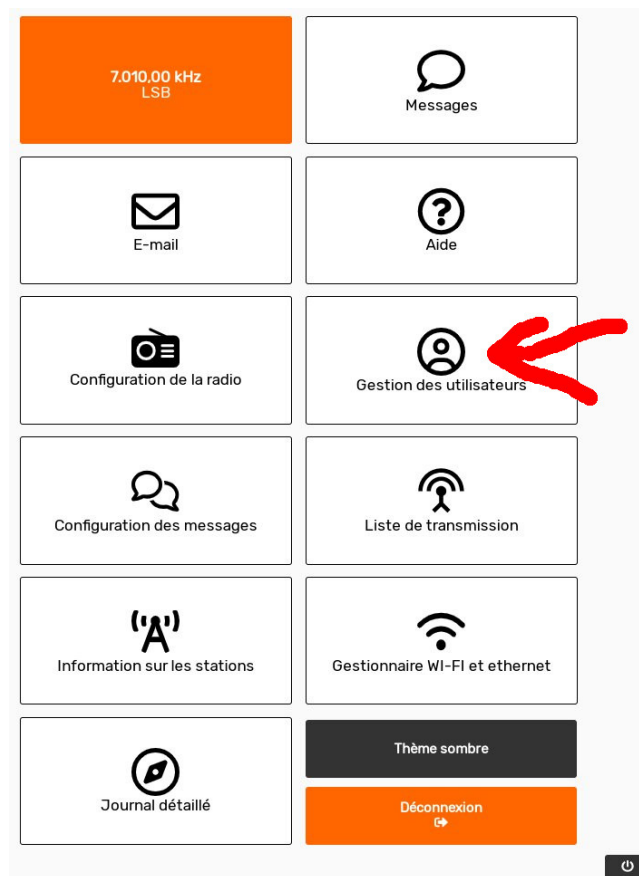


Figure 2.37: User management icon in main HERMES page.



Figure 2.38: User management icon in main HERMES page.

The in the device, first connect it to the WiFi radio network, as described in section 2.6. Then open the DeltaChat application, which available in any app store, or locally in the radio, as shown in Figure 2.39 and the, by clicking the DeltaChat icon, download links will appear, as shown in Figure 2.40.

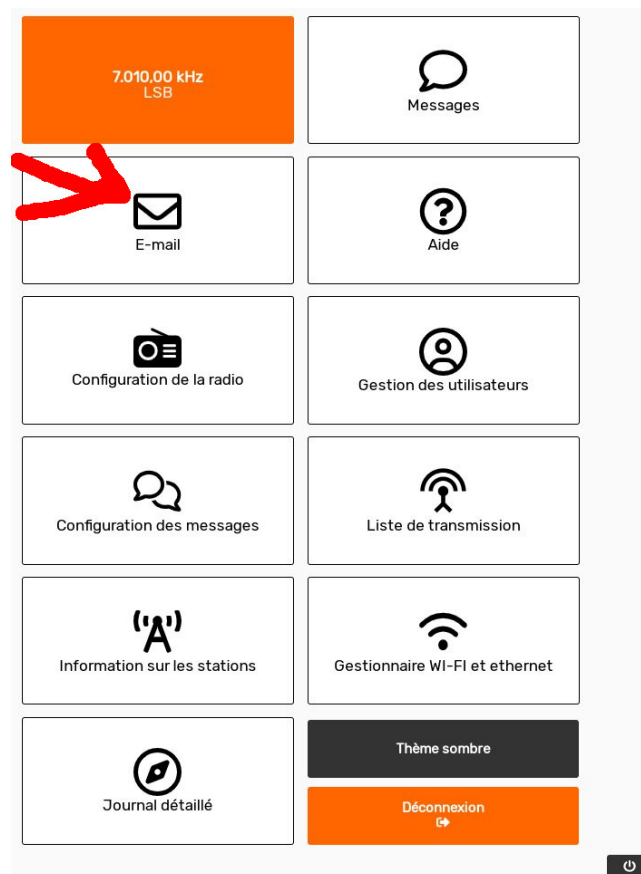


Figure 2.39: User management icon in main HERMES page.

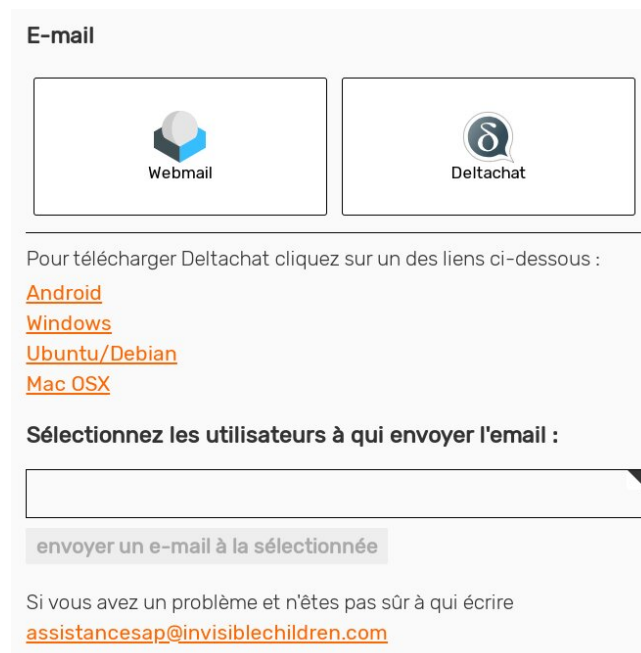


Figure 2.40: User management icon in main HERMES page.

In DeltaChat, go in “Access your server” and put the email and password created in HERMES system (see Figure 2.41). The email should look like “opérateur@irabanda.aco-connexion.org”, and is composed by the HERMES username, an “@”, and the name of the station.

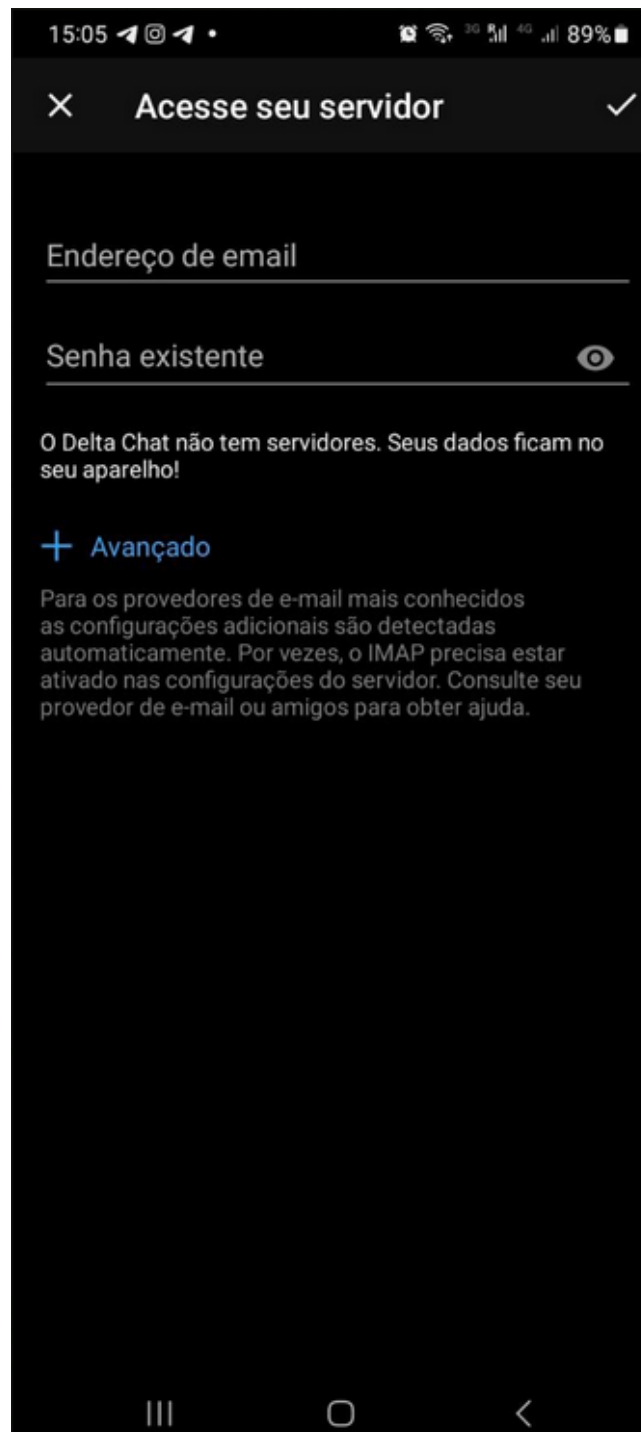


Figure 2.41: DeltaChat login screen, with email and password fields shown.

2.10 Further troubleshooting (maintainers)

This installation guide focuses on hardware setup and basic functional tests. For a detailed HERMES system architecture and software stack description (radio control via `sbitx_controller`, modem/TNC integration, `uucpd/uuport`, and gateway scheduling), see the document “**HERMES Software Description**” (`software_en.pdf`) in the HERMES documentation repository.

Useful service-level commands:

```
systemctl status sbitx uucpd
```



```
journalctl -f -u sbitx  
journalctl -f -u uucpd
```

Chapter 3

Software description (architecture and stack)

3.1 Introduction

Telecommunication in the HF Band using skywave propagation offers hundreds to thousands of kilometers coverage with relative low power and simple antennas, but it is also challenging, given the extreme limitation on throughput and very long transceiver turn-around times between tx and rx.

HERMES is a digital telecommunication system for the HF band, which uses a high-performance HF modem and UUCP (Unix-to-Unix Communication Protocol), for email transport and other ad-hoc services. UUCP is arguably the most relevant store-and-forward system to date, in use since the late 70's for software exchange, email, bulletin board systems news and even military communication. First released in Bell Labs' Unix Version 7, it is still present in all major Unixes (including Linux) and integrate seamless with email software like Postfix.

Upper network layers are composed by email software infrastructure and Web Interface for management and access to the services. HERMES network topology is flexible, but typically configured as star, with a “gateway” station located where Internet is available (in order to route data from stations located in isolated areas) and “remote” stations located at communities without access to other telecommunication systems.

3.2 HERMES Description

This section briefly introduces what HERMES system can do today. HERMES is basically composed of:

- HF transceiver and associated antenna, connected to a computer;
- Computer running the HERMES free software stack;
- The computer provides local connectivity through WiFi (or other local-area network technology)

Rhizomatica developed the HERMES hardware which embedds all the needed parts for a simple to use HF telecommunication equipment. Figure 3.1 and 3.2 contains a picture of the HERMES box. The



Figure 3.1: The HERMES hardware at a “gateway” location, a power supply (yellow), a standard HF voice radio (top right) and a wattmeter (top left).



Figure 3.2: The HERMES hardware at a “remote” location, with a battery appearing the bottom left.

HERMES box contains everything needed for joining a HERMES network, while users have access to the HERMES services over WiFi or ethernet.

HERMES uses GNU UUCP with Debian downstream patches (version 1.07-27 or greater, available at: <https://packages.debian.org/bullseye/uucp>).

Specific HERMES network stack source code, radio firmware and other system components are available at:

- Main network components and firmware: <https://github.com/Rhizomatica/hermes-net>
- REST API: <https://github.com/Rhizomatica/hermes-api>
- Web GUI: <https://github.com/Rhizomatica/hermes-gui>
- Documentation: <https://github.com/Rhizomatica/hermes-documentation>

3.3 HERMES System Architecture

This section provides an architecture-level description of HERMES as implemented by the software components referenced in Section 3.3.8. The focus is the HF data plane (store-and-forward payload transport over HF) and the radio/modem control plane that makes that possible on commodity HF transceivers.

3.3.1 Topology and station roles

HERMES is commonly deployed as a star topology network with two roles:

- **Gateway station:** has Internet access and orchestrates synchronization with remote stations.
- **Remote station:** provides local services (WiFi + web UI + email) and exchanges data with the gateway over HF.

Inter-station exchange is **asynchronous** (store-and-forward): payloads are queued locally and later transported over an HF link during scheduled communication windows (often initiated by the gateway in round-robin fashion).

3.3.2 Reference architecture (v2 / sBitx + VARA-compatible TNC + UUCP)

The reference implementation described in this document corresponds to HERMES v2 (sBitx-based radio control) and a **VARA-compatible TCP TNC interface** on localhost. In this model the station is composed of the following main processes:

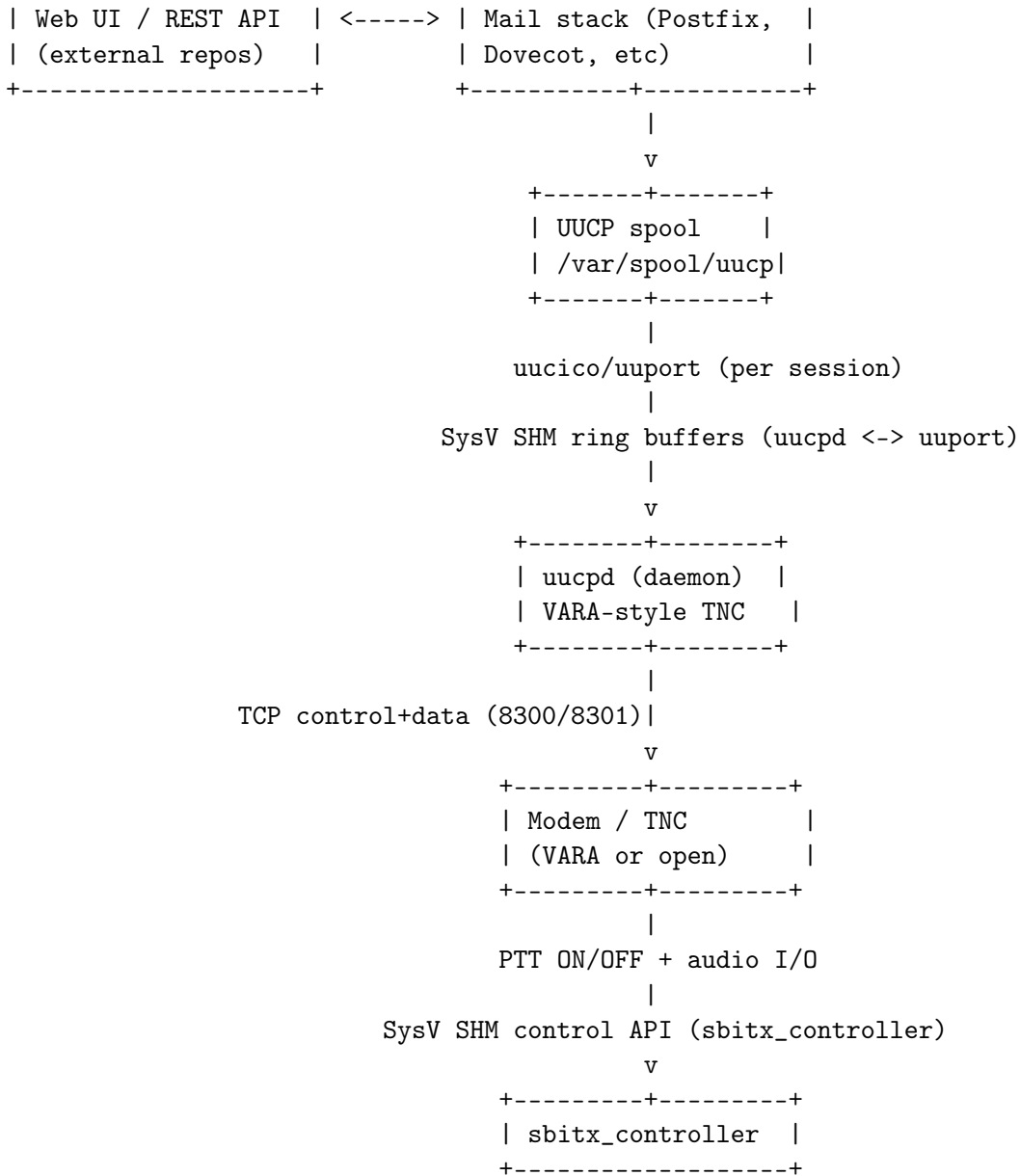
- **Radio control daemon:** `sbitx_controller` (from `hermes-net/trx_v2-userland`), started by `systemd` as `sbitx.service`. It controls RF hardware and exposes a shared-memory control/status interface used by other services.
- **Modem/TNC:** either **VARA HF** (proprietary) or an **open modem** that implements the same VARA-style TCP TNC control/data sockets (e.g. `hermes-modem`). The control socket is typically on TCP 8300 and the data socket on 8301 (base port + 1).
- **UUCP transport:** GNU UUCP (`uucico` + spool under `/var/spool/uucp`), which provides store-and-forward job management and email transport integration.
- **UUCP-to-modem bridge:** `uucpd` (daemon) + `uuport` (pipe helper) from `hermes-net/uucpd`. This is the key component that turns the modem/TNC into an UUCP “port”.
- **Gateway orchestration (gateway-only):** a scheduler such as `caller.sh` (installed as `caller.service`) that triggers `uucico -S <station>` calls according to a schedule and station list.

3.3.3 Station component diagram

User devices (WiFi/LAN)

|
HTTP(S)
v

+-----+ +-----+



3.3.4 Service orchestration (systemd)

On Debian-based HERMES images these components are typically started by systemd. Reference unit files are shipped in `hermes-net/system_services/init/`. Some helper units for the Wine-based VARA modem are shipped by `hermes-installer` under `hermes-installer/conf/x11vnc/`. The most relevant ones for the HF data plane are:

```

sbitx.service
    ExecStart=/usr/bin/sbitx_controller

# VARA HF (Wine + GUI): headless X server + VNC helper units
x11.service
    ExecStart=xvfb-run ... -n 69 -l xstartup

vnc.service
    ExecStart=x11vnc -display WAIT:69 ...

```

```

uucpd.service
    ExecStart=/usr/bin/uucpd -a 127.0.0.1 -p 8300 -r vara -o shm -f 2750p -m

caller.service    (gateway-only scheduler)
    ExecStart=/usr/bin/caller.sh

# Open modem (hermes-modem) alternative:
modem.service
    ExecStart=/usr/bin/mercury -v

uucpd-mercury.service    (typically installed as uucpd.service)
    ExecStart=/usr/bin/uucpd -a 127.0.0.1 -p 8300 -r vara -o shm

```

3.3.5 Interfaces: ports, paths and IPC

The HERMES reference architecture uses a small set of explicit interfaces. The most important ones are summarized below.

Layer	Interface	Notes
Radio control	/etc/sbitx/core.ini, /etc/sbitx/user.ini	Installed by hermes-net ; defines radio IO, profiles, etc.
Radio control IPC	SysV SHM key 66650 (controller_conn)	Command/response API used by sbitx_client and by uucpd for PTT/LED/status.
Modem/TNC	TCP base_port and base_port+1 (default 8300/8301)	VARA-style TNC interface: control socket (ASCII commands) + data socket (raw stream).
UUCP config	/etc/uucp/{ config , sys , port ,...}	config provides nodename ; sys defines peers and schedules; port uses type pipe .
UUCP spool	/var/spool/uucp	Store-and-forward queue; inspected with uustat and logs with uulog .
UUCP bridge IPC	SysV SHM keys 66664/66666/66668	Shared connector struct + in/out ring buffers used by uucpd and uuport .

3.3.6 Dynamic behavior (message and call flows)

Outbound session (gateway calls a remote station)

1. Payloads (emails/messages/files) are queued locally and end up as UUCP jobs in the UUCP spool.
2. The gateway scheduler (e.g. `caller.sh`) triggers an UUCP session: `uucico -S <remote>`.
3. `uucico` starts in master mode and invokes `uuport` (configured as a `type pipe` UUCP port).
4. `uuport` attaches to `uucpd`'s shared memory buffers and becomes `uucico`'s byte-stream "modem".
5. `uucpd` maintains a persistent TCP connection to the modem/TNC. When it sees outgoing bytes and is not connected, it issues a `CONNECT <local> <remote>` command on the TNC control socket.
6. The modem/TNC drives the RF transmit/receive turn-taking and signals `PTT ON`/`PTT OFF` on the control socket. `uucpd` translates these into actual radio keying (serial CAT or shared-memory keying via `sbitx_controller`).
7. Once connected, bytes flow bidirectionally: `uucico` ↔ `uuport` ↔ `uucpd` ↔ modem/TNC ↔ HF RF link.

Inbound session (remote answers an incoming call)

1. The modem/TNC is kept in listen mode by `uucpd` (`LISTEN ON`).
2. When a remote station calls in, the modem/TNC reports `CONNECTED ...` on the control socket.
3. `uucpd` detects this as an inbound connection and forks `/usr/sbin/uucico` in slave mode, wiring its stdin/stdout to the same shared buffers used by `uuport`.
4. The UUCP protocol runs and the spool is updated (receive jobs delivered; pending jobs may be sent back).

3.3.7 Variants

- **Open modem (hermes-modem)**: provides a VARA-style TCP TNC interface, so from `uucpd` perspective it behaves like VARA and uses the same integration path (control/data sockets, `PTT ON/OFF` messages).
- **ARDOP (legacy)**: supported by `uucpd` using its `-r ardop` mode; kept mainly for compatibility with older deployments.
- **HERMES v1 (uBitx)**: legacy radio control uses `ubitx_controller` and different service units (see `hermes-net/system_services/init`).

3.3.8 Architecture reference sources

The architecture above is grounded in the following source trees and artifacts:

- `hermes-net/uucpd` (bridge daemon `uucpd` + port helper `uuport` + modem integration code)
- `hermes-net/trx_v2-userland` (`sbitx_controller` shared-memory control API)

- `hermes-net/system_services/init` (reference systemd unit files such as `sbitx.service`, `uucpd.service`, `caller.service`)
- `hermes-modem` (open modem implementing VARA-style TCP TNC)

3.4 HERMES Software Stack

This section describes the main software components that implement the architecture described in Section 3.3.2.

HERMES deployments commonly use **VARA HF** as modem/TNC, but the system architecture is intentionally based on a VARA-style TCP TNC interface, allowing **open modem** implementations (such as `hermes-modem`) to be used as a drop-in replacement. ARDOP is supported mostly for legacy compatibility.

For the network layer, UUCP is used. UUCP is a system for asynchronous store-and-forward communication first released in Bell Labs Unix V7 in the late 1970s, and still used today in niches, like communication over HF.

The `uucpd` project (historically also referred to as `uuardopd` / `rhizo-uuardop` in older documents and code comments) was developed to integrate UUCP with HF modems/TNCs.

At a high level, the HERMES station stack is made of:

- Radio control (`sbitx_controller` and `sbitx_client`)
- Audio subsystem configuration (ALSA and related device plumbing)
- Modem/TNC (VARA HF or open modem with VARA-style TCP interface)
- UUCP (spool and `uucico` sessions)
- `uucpd` bridge (`uucpd` daemon + `uuport` helper)
- Upper services (web UI/API + email infrastructure)

3.4.1 ALSA (Audio configuration)

HERMES relies on a modem/TNC that exchanges baseband audio with the HF transceiver. On Linux, ALSA configuration is commonly used to provide stable capture/playback device mapping for the modem process (including Wine-based VARA deployments).

The reference configuration shipped in `hermes-net` is installed as `/etc/asound.conf` from `hermes-net/system_ser`

```
pcm.asymwine {
    type asym
    capture.pcm "hw:1,1"
    playback.pcm "hw:2,0"
}

pcm.!default {
    type plug
    slave.pcm "asymwine"
```



```

}

pcm.dsp {
    type plug
    slave.pcm "asymwine"
}

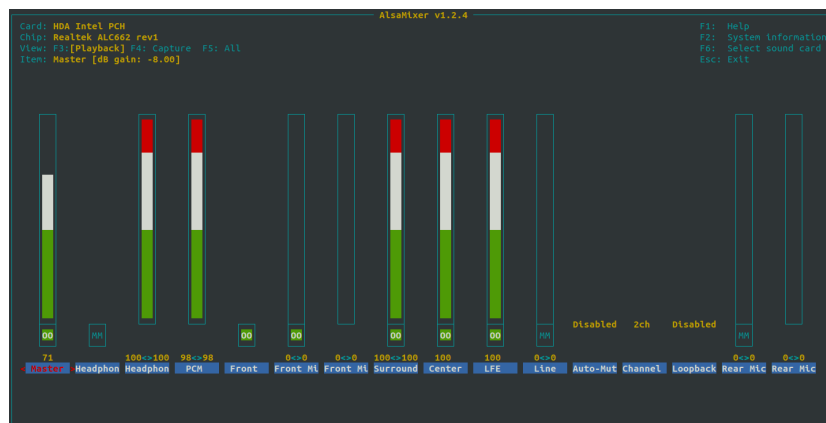
```

The actual `hw:X,Y` indices depend on your hardware. To identify them, use for example:

```

alsamixer
arecord -l
aplay -l

```



3.4.2 Radio control: `trx_v2-userland` (`sbitx_controller`)

HERMES v2 uses Rhizomatica's sBitx-based radio control stack from `hermes-net/trx_v2-userland`. It builds two main executables:

- `sbitx_controller`: a daemon that controls the radio (RF control + GPIO + I2C devices + audio DSP).
- `sbitx_client`: a CLI tool used for monitoring and setting radio parameters (frequency, mode, power, etc.).

`sbitx.service` (systemd unit)

The controller is typically started by systemd as `sbitx.service`:

```

hermes-net/system_services/init/sbitx.service
ExecStart=/usr/bin/sbitx_controller

```

Configuration files (`/etc/sbitx`)

The default configuration files are expected at:

- `/etc/sbitx/core.ini`

- `/etc/sbitx/user.ini`

They are installed by `hermes-net` (`make install`) from the `trx_v2-userland/config/` directory.

Core switches. The most relevant toggles in `core.ini` are:

- `enable_shm_control = 1` (required for `sbitx_client` and for `uucpd -o shm` integration);
- `enable_websocket = 1` (optional; used by web GUIs for status streaming).

Shared memory control API (`sbitx_client`, `uucpd`)

The controller exposes a System V shared memory segment (key 66650, type `controller_conn`, see `hermes-net/include/sbitx_io.h`). This interface is used by `sbitx_client` and also by `uucpd` when configured with `-o shm`, allowing the modem/TNC integration to key PTT (and report status such as bitrate/SNR/bytes) without direct serial/CAT access.

Websocket status API (optional)

When `enable_websocket = 1`, `sbitx_controller` starts an HTTPS/WSS server (Mongoose-based) that:

- listens on `wss://0.0.0.0:8080`;
- upgrades `/websocket` to a websocket stream (JSON status updates);
- serves static files from `/etc/sbitx/web` (installed by `hermes-net` from `trx_v2-userland/web/`).

For example, the test page is typically available at `https://<station>:8080/test.html` and connects to `wss://<station>:8080/websocket`.

TLS files are expected at:

- `/etc/ssl/certs/hermes.radio.crt`
- `/etc/ssl/private/hermes.radio.key`

Health checks

```
systemctl status sbitx
sbitx_client -c get_frequency -p 0
sbitx_client -c get_mode -p 0
```

```
# If websocket is enabled:
ss -lntp | grep 8080
journalctl -f -u sbitx
```

3.4.3 Modem configuration

VARA

VARA is a High Performance HF modem based on OFDM modulation. VARA modem is the only piece of private software in the stack.

Currently, HERMES uses VARA HF 4.3 or newer as it's standard modem. Use this protocol with your HERMES if you are looking for best performance available.

Please refer to the developer's website for download and instructions on installation and use:

<https://rosmodem.wordpress.com/>

Configuration

Edit `/opt/VARA/VARA.ini.default` to include your license key and callsign.

Headless X/VNC services (Wine VARA). In the reference Debian images, VARA HF is run under Wine and requires an X server. For headless stations, `hermes-installer` ships a minimal Xvfb + x11vnc setup:

```
hermes-installer/conf/x11vnc/x11.service
```

```
ExecStart=xvfb-run --server-args="-screen 0 1024x768x24" ... -n 69 -l xstartup
```

```
hermes-installer/conf/x11vnc/vnc.service
```

```
ExecStart=x11vnc -display WAIT:69 ...
```

The `xstartup` script (`hermes-installer/conf/xstartup`) launches `wine /opt/VARA/VARA.exe` (via `xterm`) and starts the window manager used by this session.

Integration contract (VARA-style TCP TNC). The rest of the HERMES HF stack does not interact with VARA through audio directly. Instead, `uucpd` connects to VARA's TCP interface (control socket on `base_port`, data socket on `base_port+1`; typically 8300/8301). VARA signals radio keying by sending PTT ON/PTT OFF on the control socket; `uucpd` translates that into actual PTT via serial or shared-memory radio control (`-o shm` for sBitx).

Open modem: hermes-modem (“mercury” binary)

Rhizomatica's open modem implementation is available at: <https://github.com/Rhizomatica/hermes-modem/>

It provides a VARA-style TCP TNC interface (separate control and data sockets) so it can be used with the same `uucpd` integration path described above (`uucpd -r vara ...`). In addition to ARQ links, it also provides a dedicated TCP port for broadcast data mode (default 8100).

The `hermes-modem` README documents its CLI options. In summary:

- `-p` sets the ARQ TCP base port (control `base`, data `base+1`; default 8300).
- `-b` sets the broadcast TCP port (default 8100).
- `-x` selects the audio backend (`alsa`, `pulse`, `shm`, etc.).

Systemd unit (open modem). A reference unit file is shipped as `modem.service` (see `hermes-net/system_servi`). It starts the modem directly via `/usr/bin/mercury` (installed from `hermes-modem` using `make install`). On Linux, `mercury` defaults to ALSA and uses the ALSA device `default` for capture/playback unless `-i/-o` are provided. When used, the corresponding `uucpd-mercury.service` unit depends on `modem.service`.

ARDOP

ARDOP is a SDR(software defined radio) modem compatible with both HF and VHF radio transmissions. In HERMES it is mostly a legacy/optional choice; the preferred integration target is the VARA-style TCP TNC interface described above. The protocol can operate over a wide range of data rates by automatically adjusting for optimized performance under poor multi-path conditions. Use this protocol with your HERMES if you are looking for a completely open-source and free option.

Download link: <https://github.com/DigitalHERMES/ardopc>

Configuration

The ardop binary should be in `/usr/bin/ardop`, which can be a symbolic link to `/usr/bin/ardop1ofdm`, `ardop2`, `ardopofdm`.

ARDOP service file for the ICOM IC-7100 (USB connection, PTT done over serial):

```
[Unit]
Description=ARDOP daemon

[Service]
Type=simple
ExecStart=/usr/bin/ardop 8515 -c /dev/ttyUSB0 ARDOP ARDOP -k FEFE88E01C0001FD -u FEFE88E01C0000FD
ExecStop=/usr/bin/killall -s QUIT ardop
IgnoreSIGPIPE=no
#StandardOutput=null
#StandardError=null
StandardOutput=syslog
StandardError=syslog

[Install]
WantedBy=multi-user.target
```

Service file when using a VOX based setup (eg. when using an interface like the Signalink):

```
[Unit]
Description=ARDOP daemon

[Service]
Type=simple
ExecStart=/usr/bin/ardop 8515 ARDOP ARDOP
ExecStop=/usr/bin/killall -s QUIT ardop
IgnoreSIGPIPE=no
```

```
#StandardOutput=null
#StandardError=null
StandardOutput=syslog
StandardError=syslog

[Install]
WantedBy=multi-user.target
```

Start/stop ARDOP service:

```
systemctl start ardop.service
systemctl stop ardop.service
```

See the log:

```
journalctl -f -u ardop
```

Mercury (R&D note)

The name “Mercury” is used historically in HERMES for ongoing modem R&D work. In practice, the current open modem implementation used in HERMES deployments is `hermes-modem` (previous subsection), which exposes the VARA-style TCP TNC interface expected by `uucpd`. The standalone Mercury R&D repository (when applicable) is: <https://github.com/Rhizomatica/mercury/>

3.4.4 UUCP (Network configuration)

UUCP is an acronym of Unix-to-Unix Copy. It’s a protocol for remote execution of commands and transfer of files, email and netnews between computers.

Since HERMES uses Debian GNU/Linux release Bullseye, UUCP packages are installed by default.

If you are not using Debian Bullseye, UUCP Debian package version 1.07-27 or higher should be used, for example, the version from Debian Bullseye (11): <https://packages.debian.org/bullseye/uucp>

UUCP command line examples follow. To copy a file to a remote host, the following command adds a copy job to the uucp queue (“-r” is used to not start transmission after queuing):

```
uucp -C -r -d source.xxx AM4AAB\!/var/www/html/arquivos/${nodename}/
```

Trigger the transmission of all queued jobs for host AM4AAA:

```
uucico -S AM4AAA
```

List the jobs:

```
uustat -a
```

Kill a job:

```
uustat -k job
uustat -K
```

See the log:

```
uulog
```

3.4.5 UUCPD: uucpd + uuport (UUCP to modem bridge)

UUCPD is a set of tools that allow UUCP to operate over HF links by integrating `uucico` with an HF modem/TNC. In the HERMES v2 reference architecture (Section 3.3.2) the modem/TNC is accessed through a VARA-style TCP interface (control socket on `base_port`, data socket on `base_port+1`).

The `hermes-net/uucpd` package provides two executables:

- **uucpd**: the always-on bridge daemon that keeps the modem/TNC in listen mode, initiates outgoing connects when data is available, and starts `uucico` in slave mode when an inbound call is received.
- **uuport**: the helper invoked by `uucico` when calling out (UUCP port type `pipe`). It attaches to `uucpd`'s shared memory buffers and provides a byte-stream interface to `uucico`.

Communication between `uuport` and `uucpd` uses System V shared memory (connector struct + ring buffers). When configured with `uucpd -o shm`, radio keying/status is also done through the `sbitx_controller` shared-memory interface (Section 3.4.2).

Download link: <https://github.com/Rhizomatica/hermes-net/tree/main/uucpd>

Install

To compile and install, type:

```
$ make
$ make install
```

Configuration

Port configuration example at `"/etc/uucp/port"`:

```
port HFP
type pipe
command /usr/bin/uuport
```

An alternative Port configuration if you use a patched uucp (for "\Z" support, available in "improved-pipe.patch" which was added to uucp debian package version 1.07-27), where uuport passes the callsign of the station to be called to uucpd with the uucp remote station name (allowing a single uucpd instance to be used for different remote station callsigns):

```
port HFP
type pipe
command /usr/bin/uuport -c \Z
```

Sys protocol example (tested and works fine) at `"/etc/uucp/sys"`:

```
protocol y
protocol-parameter y packet-size 512
protocol-parameter y timeout 540
chat-timeout 200
```

Sys configuration example of remote system at `"/etc/uucp/sys"` (without login prompt):

```
system remote
call-login *
call-password *
time any
port HFP
chat "" \r
```

Sys configuration example of remote system at `"/etc/uucp/sys"` (with login prompt - should call `uucpd` with `"-l"`):

```
system remote
call-login *
call-password *
time any
port HFP
chat "" \r\c ogin: \L word: \P
```

Systemd service

The reference systemd unit files shipped with `hermes-net` include:

- `hermes-net/system_services/init/sbitx.service`
- `hermes-net/system_services/init/uucpd.service`
- `hermes-net/system_services/init/caller.service` (gateway-only)

In the reference VARA deployment, the `uucpd.service` `ExecStart` is:

```
ExecStart=/usr/bin/uucpd -a 127.0.0.1 -p 8300 -r vara -o shm -f 2750p -m
```

Start/stop uucpd service

```
systemctl start uucpd.service
systemctl stop uucpd.service
```

Running uucpd (examples)

Examples of `uucpd` invocation:

```
$ uucpd -a 127.0.0.1 -c PU2BBB -p 8515 -t 60 -r ardop
$ uucpd -a 127.0.0.1 -p 8300 -r vara -o icom -s /dev/ttyUSB0 -f 2750p
$ uucpd -a 127.0.0.1 -p 8300 -r vara -o shm -f 2750p -m
```

See the log

```
journalctl -f -u uucpd
```

uucpd usage

```
-c callsign:      Station Callsign (Eg: PU2HFF)
-d remote_callsign: Remote Station Callsign (optional)
-r [ardop,vara]:  Choose modem/radio type
-a tnc_ip_address: IP address of the TNC
-p tcp_base_port: TCP base port of the TNC. ARDOP uses ports tcp_base_port and tcp_base_port
-t timeout:      Time to wait before disconnect when idling (ARDOP ONLY)
-f features:      Enable/Disable features.
                  Supported features ARDOP: ofdm, noofdm (default: ofdm)
                  Supported features VARA, BW mode: 500, 2300 or 2750 (default: 2300)
                  Supported features VARA, P2P mode: "p" to enable (eg. 2300p)
-s serial_device: Set the serial device file path for keying the radio (VARA ONLY)
-l:              Tell UUCICO to ask login prompt (default: disabled)
-F:              Enable UUCP handshake fast-track (requires both ends; no-login chat)
-m:              Enable shared memory messaging parameter to uucp (requires patched uucp)
-o [icom,ubitx,shm]: Sets radio type (supported: icom, ubitx or shm).
-h:              Prints this help
```

UUPORT Usage

```
-c system_name:  Name of the remote system (default is don't change).
-e logfile.txt:  Log file (default is stderr).
-h:              Prints this help
```

3.4.6 User Web Interface

HERMES provides a web-based interface for sending messages, files and emails. It also allows for user management and radio configuration, among other features. Current implementation also supports symmetric cryptography and audio/image compression.

The web app is composed of the front-end interface and a REST API.

Hermes-GUI

Hermes-GUI is the front-end interface built with Angular.js.

Source code is located at: (<https://github.com/Rhizomatica/hermes-gui/>).

The project was generated with Angular CLI version 13.2.6. Angular: 13.2.6 Angular CLI: 13.2.6
Node: 12.22.5 Package Manager: npm 8.5.2

Requirements of the user interface:

- ImageMagic: for image manipulation
- VVC: Best free software low bitrate visual content compression codec: <https://github.com/fraunhoferhhi/>
- LPCNet: Best free software audio compression codec: <https://github.com/xiph/LPCNet>
- GnuPG: For cryptography
- hostapd: WiFi AP mode software

The Web interface can be accessed by typing any address in a browser connected to the WiFi (set the DNS accordingly) or simply 10.0.0.1.

Install node, npm

Install node preferably(V12.22.5) and npm (node package manager) in your distro

Run 'npm install' inside the project path

Development server

Configure .env file with your parameters and run npx ts-node setEnv.ts to set .env values in environment.ts.

Run ng serve --configuration=en for a dev server in english, you can change the language to spanish (ng serve --configuration=es) or portuguese (ng serve --configuration=pt) if you wish. Navigate to <http://localhost:4200/>. The app will automatically reload if you change any of the source files.

Code scaffolding

Run ng generate component component-name to generate a new component. You can also use ng generate directive|pipe|service|class|guard|interface|enum|module.

Running unit tests

Run ng test to execute the unit tests via Karma.

Running end-to-end tests

Run ng e2e to execute the end-to-end tests via Protractor.

Further help

To get more help on the Angular CLI use ng help or go check out the Angular CLI README.

Interface Contents

Inside the folder src you will find the angular templates for the system's interface.

On style.less there are the styling code, written in less.

Inside the app folder, you will find the template components. The app-routing module.ts file is responsible for linking the templates to url addresses. and the app.component, the application root template.

For every system component, there will be a folder with the component.html, component.less, and component.ts. On the .html file there is the html angular template, and on the component.ts you will find typescript code related to that component.

Also inside the app folder, there is the _services folder, where you can find common services shared between the components to access the [station-api] (<https://github.com/DigitalHERMES/station-api>).

On assets folder you find links to svgs and image files used on the interface design.

On the locale folder there are the xlf files for translation. For the translations, we are using i18n angular module. To generate translations, you need to use `ng extract-i18n --output-path src/locale/` to generate the messages.xlf file and then `xliffmerge --profile xliffmerge.json pt es` to transpose the new data to both messages.es.xlf and messages.pt.xlf, where you can input the new translation tokens.

Hermes-API

Hermes-API is a REST api for use on Hermes stations to exchange messages between them. It uses Lumen PHP Framework and composer to manage its own dependencies.

Source code is located at: (<https://github.com/Rhizomatica/hermes-api>).

Server Requirements:

- Web server
- PHP ≥ 7.3
- OpenSSL PHP Extension
- PDO PHP Extension
- Mbstring PHP Extension
- SQLite DataBase

Installation settings

Setup your installation settings creating a .env file from .env.example

Run Composer

Composer is a dependency manager for PHP.

Run composer in your folder to install the dependencies and update relevant packages:

```
COMPOSER_ALLOW_SUPERUSER=1 composer install
COMPOSER_ALLOW_SUPERUSER=1 composer update
```

Copy hermes-api to /var/www/station-api/

Go to your hermes-api folder and run:

```
cp -rT . /var/www/station-api/
```

GNU Privacy Guard

Create the folder for GNU Privacy Guard and change owner:

```
mkdir -p /var/www/.gnupg
chmod 700 /var/www/.gnupg
chown www-data:www-data /var/www/.gnupg
```

Setup public folders for UUCP

Create and give permissions to inbox/outbox, downloads and temp folders to uucp public:

```
mkdir -p /var/www/station-api/storage/app/inbox
chmod 777 /var/www/station-api/storage/app/inbox
mkdir -p /var/www/station-api/storage/app/outbox
chmod 777 /var/www/station-api/storage/app/outbox
mkdir -p /var/www/station-api/storage/app/downloads
chmod 777 /var/www/station-api/storage/app/downloads
mkdir -p /var/www/station-api/storage/app/tmp
chmod 777 /var/www/station-api/storage/app/tmp
mkdir -p /var/www/station-api/storage/app/uploads
chmod 777 /var/www/station-api/storage/app/uploads
```

The structure created is used by the app as follows: /uploads (Files of outgoing messages) /downloads (Files generated from the inbox received messages) /inbox (incoming hermes message packs) /outbox (hermes message pack for deliver) /tmp (tmp files)

Link folders

```
ln -sf /var/www/station-api/public/ /var/www/html/api
```

Change owner of the app folder

```
chown -R www-data:www-data /var/www/station-api/
```

create database.sqlite file in database folder

To start a fresh database: php artisan migrate:refresh --seed

Database setup

Create a database named hermes and user with all privileges:

```
echo "CREATE USER hermes IDENTIFIED BY 'db_hermes'; " > sql_commands.sql
echo "CREATE DATABASE hermes;" >> sql_commands.sql
echo "GRANT ALL PRIVILEGES ON hermes.* TO hermes;" >> sql_commands.sql

mysql < sql_commands.sql
```

Then go to your hermes-api folder and run Artisan for the migrations:

```
cd /var/www/station-api/
php artisan migrate
php artisan db:seed
```

Running on port 8000:

```
> php -S localhost:8000 -t public
```

Hermes Message Pack

Hermes Message Pack is a tar gzipped file named .hmp

Chapter 4

Hardware description

4.1 HERMES Hardware

The HERMES setup is composed by different elements. The HF transceiver is responsible for receiving and transmitting signals over-the-air, allowing the exchange of digital data. The transceiver is necessarily connected to an antenna and a computer. Also a power source is needed. The list of main components for HERMES are the following:

- HF Radio Transceiver
- Antenna
- Mini-Computer
- Power source

Each part has its own dedicated subsection, and also a section with important tools to have in order to properly adjust the HF station is present.

4.1.1 HF Radio Transceiver

Many options for HF transceiver are available, but in order to allow easy operation and better integration of the overall HERMES system, Rhizomatica developed its own HF transceiver, based on the μ Bitx v6 board (See: <https://www.hfsignals.com/index.php/ubitx/>).

If not using Rhizomatica's transceiver, we consider best options for HF radio transceiver the ones which have a USB (Universal Serial Bus) port and internal analog/digital converters, which provide seamless and error-free connection between the HF transceiver and a computer. Taking in consideration the cost of the transceivers, usually the most affordable ones are made for the amateur (ham) radio market. Ham radio transceivers are usually blocked to transmit only inside the ham radio allocated bands, and need to be modified to transmit in all the HF bands (See example here: <https://radioaficion.com/cms/ic-7100-marscap-modification/>).

In order to allow HERMES modem to work, a pass-band of at least 2.7kHz is recommended in SSB mode (the mode typically used for data and voice transmissions in HF).

Among this category we list as examples (supported frequency bands between parenthesis):

- ICOM IC-7100 (HF / VHF / UHF)
- ICOM IC-7300 (HF)
- Yaesu FT-991A (HF)

Other HF transceivers can also be used for digital telecommunications, but need an external interface to connect the radio to a computer. Examples of transceivers in the category are:

- ICOM IC-7000 (HF / VHF / UHF)
- ICOM IC-78 (HF)
- ICOM IC-718 (HF)
- Vertex VX-1700 (HF)
- Alinco DX-SR8T (HF)
- Yaesu FT-857D (HF / VHF / UHF)
- Yaesu FT-891 (HF)

If an external interface is needed to connect a transceiver to computer, be aware that an adjust procedure to identify the correct audio levels to correctly drive the transceiver and to optimally receive the audio needs to be done. With some transceivers like the ICOM IC-78 and IC-718 there is a need to set the MIC gain level to 0 when transmitting using its ACC connector. Different transceivers have different quirks and issues, so be aware to always carry tests before taking any equipment to the field. This is one of the reasons Rhizomatica took to path to develop its own transceiver.

Examples of external interfaces which support all transceiver models (PC connection between parenthesis):

- Tigertronics Signalink USB (USB)
- DigiMaster MiniProSC (USB)
- West Mountain Radio RIGblaster (USB and Bluetooth)

4.1.2 Antenna

The antenna is a very special part of the system. If not well tuned or well installed, nothing works. We recommend the use of the appropriate antenna for the desired coverage. For Near Vertical Incidence Skywave (NVIS), which provides up to 600km radius of coverage, we recommend a simple quarter wavelength antenna with a balun for impedance matching. If a portable antenna is needed, the Bud-dipole antenna is a good option, but with worse performance then a simple 1/4 wavelength dipole (See <https://www.buddipole.com/>). Frequency bands below 7MHz are recommended for NVIS operation.

For wider coverage and more-or-less omnidirectional radiation pattern, the installation of a dipole as inverted V is desired.

4.1.3 Mini-computer

The mini-computer will host the HF modem application, the network stack and services. In the minimal setup, the computer will run the HF modem application and a Web server which provides access to the HF telecommunication system services over a WiFi network. A Raspberry Pi 4 has enough processing power and memory for this use case. For a proper HERMES installation, with full email services, automatic media compression and WiFi access point, the following configuration is recommended:

- Intel-based computer (4th generation or better);
- 4GB of RAM;
- 120 GB of storage (PCIe NVMe highly recommended) is needed;
- WiFi adapter with support to access point mode (M.2 type recommended);
- 12V to ATX power supply (for off-the-grid use cases).

Optionally, instead of WiFi for user access to services, HERMES can host a mobile network access point, for example, using GSM or LTE technologies.

4.1.4 Power source

A common HF transceiver uses a 12V voltage input and at least 1A current in receiving mode, and peaks up to 25A when transmitting. So it is important to have a charge controller which can provide at least 30A, in order to allow some headroom for fluctuations of consumption.

As a example scenario, take a transceiver using 20A when transmitting full power, 1A while listening. Lets assume we want to transmit for 5 minutes every 24 hours and listen all the time when we're not transmitting.

The daily power budget of the system (@12V) is:

HF rx: $1A * 24h = 24Ah$

HF tx: $20A * 5/60h = 100/60 = 1.7Ah$

mini-pc: $0.5A(1.2A@5V) * 24h = 12Ah$

So total daily battery consumption would be $24 + 1.7 + 12 = 37.7Ah$

Let's say we want to discharge our batteries 15% on a normal day, so this load would represent a battery capacity of $100/15 * 37.7 = 251.3Ah$.

We'd probably also want some spare capacity, say, an extra day of using the system when there's no sun. (solar panels are really bad at producing output on cloudy days: expect 8%-20% of normal output). So we'll either have to add more batteries or stop listening all day (nearly all our power consumption is for listening, not talking). for the solar panels recharging the battery, it is a good practice to size things at somewhere around 2 times the normal load every day, so that the system recharges in about 1 day after 1 day of operating without sun, plus add around 15%-20% for inefficiencies.

The solar maps for the Amazon region show a yearly average insolation of about $4.0kWh/m^2/d$. Based on this,

$$4h * C_{panel}(W) = 37.7Ah * 12.5V * 2.2$$

$$C_{panel}(W) = 2.2 * (37.7Ah * 12.5V) / (4h)$$

$$= 259.2W$$

So there is a need for around 250W to 300W of solar panels.

Another good practice would be to double the battery capacity for this scenario to make the system workable for more than 1 sunless day without discharging the batteries too deeply.

Recommended power related parts:

- 250Ah stationary battery
- One 250W or 300W solar panel, or two 250W panels
- Charge controller which can output at least 30A of current. 40A recommended.

In the case of using the HF transceiver connected to power grid, a AC/DC power supply is needed, with 12V or 13.8V output. A couple of options for the AC/DC power supply:

- At least 30A transformer-based power supply at 12V or 13.8V.
- Low cost switched power supply at 12V or 13.8V marked as 50A or more.

4.1.5 Essential tools

Essential tools to have for basic radio and antenna testing:

- Wattmeter with reflected and forward power readings for HF, which supports at least 100W
- Multimeter
- 50ohm dummy load for at least 100W

Chapter 5

sBitx notes

5.1 Introduction

This document address all the software details needed for the sBitx hardware bringup.

5.2 sBitx Hardware

5.2.1 Power consumption

5.3 Hardware interfaces to the computer (RPi 4)

5.3.1 Control I/O

I2C RTC

I2C Si5351

I2C ATtiny85 (FWD and REF)

Digital pins setup for knobs control

5.3.2 Signal I/O

I2S Wolfson wm8371

5.4 Linux setup for sBitx operation

5.4.1 Pin attribution setup for the RPi 4

5.4.2 Audio - ALSA subsystem

5.4.3 Ashhar's sbitx reference implementation

power setup

5.4.4 bridge setup

5.4.5 I2C bit-banging

5.4.6 RTC setup